Born to Lose: The Political and Fiscal Arithmetic of Argentine Social Security

Manuel Macera and Andrés Neumeyer
Universidad Torcuato Di Tella

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Abstract[*]

Argentina's fertility rate fell from 2.35 in 2015 to 1.5 in 2023. In the implied demographic steady state, the old-age dependency ratio more than doubles. We study the intergenerational incidence, fiscal sustainability, and politics of Argentina's pay-as-you-go pension system (Sistema Integrado Previsional Argentino, or SIPA), in an overlapping-generations model with population dynamics driven by 2023 age-specific fertility and survival rates. Cohort welfare turns on one sufficient statistic: the present value of remaining retirement benefits per peso of remaining contributions. At a 6% market return, a balanced-budget SIPA returns 43 cents per peso to an entering worker under today's age distribution or 18 cents under the stationary-state distribution implied by current fertility; the debt-financed status quo returns 68 cents and exhausts its borrowing capacity around 2090. Restoring balance requires pension revenues to rise from 4.3% of GDP to 16.5%, or the average effective replacement ratio to fall from 58% to 15%. Because the sufficient statistic rises with age, each policy has a break-even cohort: 29 years old under the status quo; 37 under a balanced budget today; 50 in steady state. Argentina's 2023 median voter, under compulsory voting, was 41. SIPA redistributes from young to old, and an aging electorate sustains it.

Keywords: social security, demographic transition, fiscal sustainability, political economy of reform, overlapping generations, Argentina

JEL Classification: D72, E62, H55, H62, H63, J11, J18, J26

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1. Introduction

Argentina's total fertility rate fell from 2.35 children per woman in 2015 to 1.5 in 2023; the old-age dependency ratio rose from below 0.10 in 1950 to 0.24 in 2023 and is projected to reach about 0.6 in steady state; and the natural population growth rate is on track to turn negative by 2030. This paper examines the impact of population aging on three aspects of its social security system: intergenerational redistribution, fiscal sustainability, and the politics of reform.

The literature on Argentina's social security system has focused on the challenges posed by labor market informality: A small number of formal workers support a large number of retirees who never contributed to the system. We abstract from informality to focus on demography and assess whether, even if the system worked perfectly and had no informality, it would be worth keeping.

Our main finding is that under contemporary Argentine demographics, Argentina's pay-as-you-go (PAYG) pension system, the Sistema Integrado Previsional Argentino (SIPA),[1] resembles a *compulsory Ponzi scheme* in Samuelson's (1958) sense: Current workers transfer resources to current retirees, expecting that future cohorts will, in turn, transfer resources to them. With an aging population, this is a bad bargain for the generation entering the labor market, as the implicit return on their contributions is negative and far below the market return on savings. A voluntary Ponzi scheme with these returns would unravel, defaulting on previous investors. PAYG social security does not, because the median voter benefits from its redistribution and makes the young population's contributions compulsory.

We perform our quantitative exercises in the simplest setting. We embed a stylized Argentine institutional setting in an overlapping-generations model with uncertain lifetimes, age-dependent fertility and survival, and complete markets in survival-contingent claims. The economy is small and open; it faces an exogenous interest rate; and labor supply is inelastic. By construction, wages and aggregate output are essentially exogenous, and the only role of the pension system is to redistribute consumption across cohorts. This *endowment-economy* structure isolates how demographic change alone reshapes the intergenerational incidence of PAYG pensions, extending Samuelson's (1975) classic analysis. The model delivers closed-form expressions for cohort welfare as a function of a single sufficient statistic: the return on remaining contributions $\hat{r}_{a,t}$—that is, the ratio of the expected present value of remaining benefits to the expected present value of remaining contributions for the representative agent of age a in period t. This statistic links demographics, policy parameters, and individual survival risk, and it shows why the same arrangement can lower the welfare of the young and raise that of near-retirees.

In our benchmark calibration, with a market return of $r = 6\%$ and productivity growth of

[1] Administered by the Administración Nacional de la Seguridad Social.

$\gamma = 2\%$, a *balanced-budget* SIPA gives a labor market entrant who contributes throughout their working life an implicit return of $\hat{r}_0 = 0.43$ under the 2023 age distribution (a loss of 57% of contributions in present value) and $\hat{r}_0 = 0.18$ (a loss of 82%) under the stationary age distribution implied by 2023 fertility and survival rates; the return along the demographic transition lies between these bounds. Because workers treat past contributions as sunk, the return on remaining contributions rises with age, so there is a cutoff age below which voters oppose the system and above which they favor it: 37 under the 2023 age distribution and 50 in the demographic steady state (Table 3). In both cases the fraction of the population above the cutoff exceeds one-half. Argentina's 2023 median voter was 41, above the 2023 cutoff of 37. A lower benchmark return, $r = 4\%$, softens but does not reverse these findings.

The balanced-budget exercises let social security parameters adjust to demography every period. We also evaluate a frozen status quo, calibrating SIPA revenues and outlays to their observed 2023 values of 4.3% and 5.6% of GDP (AFIP, 2024; Subsecretaría de Seguridad Social, 2024). As the population ages, the deficit widens steadily and the cumulative pension debt crosses the natural borrowing limit around 2090 (Figure 4); even this unsustainable system returns only 68 cents per peso labor market entrants contribute ($\hat{r}_0 = 0.68$). Restoring period-by-period balance with a single instrument is quantitatively dramatic: Adjusting contributions alone, pension revenues must reach roughly 16.5% of GDP by the end of the century; adjusting benefits alone, the average effective replacement ratio must fall from 58% to about 15% by 2100 (Figure 5). Extending the retirement age helps but is insufficient.

Finding a political coalition to close the fiscal gap is challenging. Because the status quo consumes beyond the economy's means, fiscal adjustment makes every generation worse off; the question is who loses most. Future cohorts lose far more under contribution adjustment (losses approach 18% of permanent consumption) than under benefit adjustment (3%–4%). Cohorts alive today fare worse under benefit adjustment, which cuts the pensions of people already retired and hits workers in early middle age hardest, with losses of about 10% of permanent consumption. The median voter accordingly prefers closing the gap by taxing the young and the unborn.

Our model captures informality only through the effective contribution rate, which is calibrated to observed SIPA revenues: The coverage gap lowers the average rate paid by the whole working-age population instead of concentrating contributions on the formal minority—the best case for the formal young. In reality, informal workers are roughly 40% of employment (INDEC, 2025) and disproportionately young (Appendix B), so a formal contributor's return per peso is below the blended $\hat{r}_0$ we report; Section 6 discusses why bringing them into the system is fiscally attractive but individually irrational.

1.1 Related Literature

Our paper quantitatively explores the impact of recent demographic changes on three interrelated questions: whether a PAYG pension system is fiscally sustainable as the population ages, how much it redistributes across generations, and why a system that expropriates the young survives majority voting. We organize this section around these three questions and close with the literature on the Argentine system.

The starting point is Samuelson (1958): In an overlapping-generations economy a PAYG transfer from workers to retirees pays each cohort a "biological" return equal to the growth rate of the wage bill, $(1+n)(1+\gamma)-1$. Samuelson (1975) and Aaron (1966) show that the transfer raises the welfare of every generation if and only if this return exceeds the market interest rate, r, a result known as the "social insurance paradox." We build on the work of Balasko and Shell (1980), who show that in an overlapping-generations economy with finite wealth at equilibrium prices, a competitive allocation is Pareto optimal, and that of Geanakoplos et al. (1998), who develop the present-value accounting that underlies our zero-sum identity. Browning (1975) is the earliest median-voter model of social insurance, and it introduces the main elements of our model with a cutoff age below which voters oppose a larger system and above which they favor it. Previous quantitative papers trace how population aging shifts the pivotal voter to make reform politically hard in the United States (Galasso, 1999; Conesa and Krueger, 1999; Bohn, 1999). Gonzalez-Eiras and Niepelt (2008) characterize the Markov-perfect politico-economic equilibrium in which aging raises pension spending and taxes.

In all these models, intergenerational redistribution drives the political conflict. Its measurement is the subject of generational accounting. Auerbach et al. (1991, 1994) and Kotlikoff (1992) measure it. Auerbach et al. (1999) apply their method to countries around the world, and their volume contains a chapter on Argentina, Altamiranda (1999), which finds a substantial generational imbalance under the Convertibility Plan, with the transition cost of the 1994 pension reform being a major driver.

The quantitative literature on the sustainability of PAYG systems begins with Auerbach and Kotlikoff (1987), whose large-scale overlapping-generations model made it possible to simulate the transition path of taxes, benefits, and capital under alternative pension rules. De Nardi et al. (1999) project the US system under the demographic transition and document the large welfare losses that the required tax increases impose on workers and on cohorts not yet born. A sequence of papers then quantifies the adjustment that aging requires—for example, Kotlikoff et al. (2007), Nishiyama and Smetters (2007), Attanasio et al. (2007), Krueger and Ludwig (2007), Börsch-Supan et al. (2006), Conesa and Garriga (2008), İmrohoroğlu and Kitao (2012), McGrattan and Prescott (2017), and Bohn (1999, 2001). Latin America has its own applied literature on the fiscal consequences of aging. Cotlear (2011) asks whether the region is ready for aging; Bosch et al.

(2013), Altamirano Montoya et al. (2018), and Álvarez et al. (2020) project pension spending and coverage across the region and stress that contributory systems whose coverage rests on formal employment will underperform in labor markets with widespread informality; and Bravo and Uthoff (1999) estimate the implicit pension debt and the transition costs of the region's funded reforms.

Researchers focusing on Argentina's pensions have studied the system mainly as a sequence of reforms and reversals. The 1994 introduction of individual pension accounts, which switched from PAYG to a fully funded system with defined contributions, and its macroeconomic effects are studied in FIEL (1998), Rofman (2001), and Bertranou (2003), and in Cetrángolo and Grushka (2004), who emphasize how the cost of the transition contributed to the 2001 crisis. The 2005–8 moratoria and the 2008 nationalization are analyzed by Rofman et al. (2010) and Mesa-Lago (2009). Bertranou et al. (2012, 2018), Bertranou et al. (2019), and Cetrángolo and Grushka (2020) provide the institutional and fiscal accounting of the current system—fragmentation across regimes, the shift to general-revenue financing, and spending that is high for the country's demography—and Rofman (2025) and Mera (2023) quantify the cost of the special regimes and moratoria. On coverage and informality, Rofman and Oliveri (2012) and Rofman et al. (2015) document that the moratoria raised coverage among the elderly from about two-thirds to about 90% at a cost of roughly 2.5% of GDP. On the demographic transition itself, Gragnolati et al. (2015) use national transfer accounts to show that Argentina's demographic window closes around 2035–40 and that pensions are the channel through which aging turns intergenerational transfers against workers; Apella (2022) provides the official long-run benchmark, with pension spending rising from about 8.5% to 12% of GDP by 2050 and 16% by 2100 as the support ratio falls from three to one. These projections are the closest counterparts to our status quo path.

To the best of our knowledge, no study of the Argentine system estimates the return on contributions and the welfare change to every cohort along the demographic transition and places the resulting break-even ages against the age distribution of voters in one calibrated model.

The remainder of the paper is organized as follows. Section 2 presents the multiperiod overlapping-generations model and the demographic block. Section 3 derives closed-form expressions for the intergenerational distributional impact of social security. Section 3.3 shows that every fiscally feasible policy is Pareto optimal. Section 4 brings the model to Argentine data and evaluates the distributional consequences of reforms. Section 5 compares the break-even age with the age distribution of voters, nationally and across provinces. Section 6 discusses interpretation, limitations, and extensions. The appendixes document the institutional details of SIPA that discipline our calibration (Appendix A), the size and age distribution of the informal sector (Appendix B), population dynamics (Appendix C), the certainty-equivalent stock market return behind our benchmark interest rate (Appendix D), the proofs for Section 3.3 (Appendix E), and the disutility-adjusted welfare change (Appendix F).

2. A Framework to Analyze Intergenerational Distributive Conflict Under Social Security

Consider a small open economy facing an exogenous international interest rate r, inhabited by multiperiod-lived, overlapping generations, with an inelastic labor supply, a life-cycle income profile, uncertain lifespans, no intergenerational altruism, and access to complete financial markets.

2.1 Demographics

Agents are born with age $h = 0$, live for at most $T+1$ periods, and supply labor up to age $R < T$. An agent of age h can offer θ_h efficient units of labor. We use $\mathscr{P}_{h,t}$ to denote the mass of agents of age h in period t. We denote the total population by $\mathscr{P}_t = \sum_{h=0}^{T} \mathscr{P}_{h,t}$, the active population by $N_t = \sum_{h=0}^{R} \mathscr{P}_{h,t}$, the dependent population by $D_t = \sum_{h=R+1}^{T} \mathscr{P}_{h,t}$, and the aggregate labor force by $L_t = \sum_{h=0}^{R} \theta_h \mathscr{P}_{h,t}$.

The age distribution of the population is determined by vectors of age-dependent fertility rates and survival probabilities. It evolves following the difference equations

$$\mathscr{P}_{0,t+1} = \frac{1}{2} \sum_{h=0}^{T} \phi_{h,t} \mathscr{P}_{h,t} \tag{1a}$$

$$\mathscr{P}_{h+1,t+1} = \frac{S_{h+1,t}}{S_{h,t}} \mathscr{P}_{h,t} \qquad \text{for } h = 0, \ldots, T-1, \tag{1b}$$

where $\phi_{h,t}$ are the age-dependent average number of children per woman aged h years in period t, $S_{h,t}$ is the unconditional probability of reaching age h, and T denotes the maximum lifespan so that $S_{h,t} = 0$ for $h > T$.

Equation (1a), the first demographic equation, represents total births in year t under the assumption of a sex birth ratio of one-half and equal survival probabilities across sexes. Population dynamics after birth are determined by survival probabilities and are described by the difference equations in (1b). Population dynamics depend on the probability of survival to age $h+1$, conditional on having survived to age h—that is, $\frac{S_{h+1,t}}{S_{h,t}}$. Probability $S_{0,t}$ in (1b) is unconditional $(S_{-1,t} = 1)$.

We can derive the fundamental equation of demography from the system of difference equations (1); writing the population growth rate n_t as the difference between the birth rate and the death rate,

$$n_t = \underbrace{\frac{\frac{1}{2}\sum_{h=0}^{T} \phi_{h,t} \mathscr{P}_{h,t}}{\mathscr{P}_t}}_{\text{birth rate}} - \underbrace{\frac{(1 - S_{0,t})\mathscr{P}_{0,t} + (1 - \frac{S_{1,t}}{S_{0,t}})\mathscr{P}_{1,t} + \ldots + (1 - \frac{S_{T-1,t}}{S_{T-2,t}})\mathscr{P}_{T-1,t} + \mathscr{P}_T}{\mathscr{P}_t}}_{\text{death rate}}.$$

Let $\pi_t \in \mathbb{R}^{T+1}$ be the age distribution of the population at t, with elements $\pi_{h,t} = \mathscr{P}_{h,t}/\mathscr{P}_t$. The age distribution of the active population is $\pi^N_{h,t} = \mathscr{P}_{h,t}/N_t$ for $h = 0,\ldots,R$. Analogously, the age distribution of the retired population is $\pi^D_{h,t} = \mathscr{P}_{h,t}/D_t$ for $h = R+1,\ldots,T$. The age distribution of the population follows a system of first-order difference equations derived from (1) with initial condition π_0, which we describe in Appendix C.

2.2 Consumer Behavior

Agents maximize the expected discounted value of utility, face the risk of an uncertain lifespan, are not altruistic, and supply their available labor inelastically.

Consumer preferences at age a and time t are

$$\mathbf{E}_t\left[\sum_{h=a}^{T}\beta^{h-a}u(c_{h,t+h-a})\right] = \sum_{h=0}^{T-a}\beta^h\frac{S_{a+h,t+h}}{S_{a,t}}u(c_{a+h,t+h}), \tag{2}$$

where $c_{h,t}$ denotes consumption at age h and time t. Consumption while agents are alive is certain, and agents do not value consumption when they are dead.

In any period $t+h$, each agent can purchase securities that pay $1+r$ in the following period conditional on survival, at price $q_{h,t}$. Their budget constraints during the active ($a \le h \le R$) and retirement ($h > R$) periods are

$$c_{h,t+h} + q_{h,t+h}b_{h,t+h+1} \le (1-\tau_{t+h})W_{t+h}\theta_h + (1+r)b_{h-1,t+h} \qquad \text{for } h \le R \tag{3a}$$

$$c_{h,t+h} + q_{h,t+h}b_{h,t+h+1} \le p_{t+R+1-a,t+h} + (1+r)b_{h-1,t+h} \qquad \text{for } h \ge \max\{a, R+1\}, \tag{3b}$$

with initial condition $b_{-1,t} = 0$ for a newborn cohort ($a = 0$); for cohorts already alive at t, $b_{a-1,t}$ is not an initial condition but an equilibrium object determined by the cohort's past optimization and is generally nonzero. While active, agents earn labor income by supplying θ_h efficient units of labor at wage W_{t+h}, net of the social security tax τ_{t+h}. Agents who retire at date $t+R+1-a$ receive the pension $p_{t+R+1-a,t+h}$ thereafter. Using the actuarially fair bond prices $q_{h,t} = S_{h+1,t+1}/S_{h,t}$ above and summing the budget constraints over the life cycle yields the Arrow-Debreu intertemporal budget constraint

$$\begin{aligned}\sum_{h=a}^{T}\frac{S_{h,t+h-a}/S_{a,t}}{(1+r)^{h-a}}c_{h,t+h-a} \le \sum_{h=a}^{R}\frac{S_{h,t+h-a}/S_{a,t}}{(1+r)^{h-a}}(1-\tau_{t+h-a})\theta_h W_{t+h-a} \\ + \sum_{h=R+1}^{T}\frac{S_{h,t+h-a}/S_{a,t}}{(1+r)^{h-a}}p_{t+R+1-a,t+h-a}.\end{aligned} \tag{4}$$

The consumer's first-order conditions deliver the standard Euler equation,

$$u'(c_{h,t+h}) = \beta(1+r)u'(c_{h+1,t+h+1}), \tag{5}$$

and the intertemporal budget constraint (4) pins down the consumption profile.

It is useful to define the present value of wage income for the remainder of life at age a and time t ($\mathscr{W}_{a,t}$), the disposable present value of remaining wages ($\mathscr{W}^d_{a,t}$), the average social security tax rate ($\hat{\tau}_{a,t}$), and the present value of retirement income at age a and time t ($\mathscr{R}_{a,t}$) as

$$\mathscr{W}_{a,t} = \sum_{h=0}^{R-a} \frac{S_{h+a,t+h}/S_{a,t}}{(1+r)^h}(1+\gamma)^h \theta_{h+a} W_t \tag{6a}$$

$$\mathscr{W}^d_{a,t} = (1-\hat{\tau}_{a,t})\,\mathscr{W}_{a,t}, \text{ where } \hat{\tau}_{a,t} = \sum_{h=0}^{R-a} \frac{\frac{S_{h+a,t+h}/S_{a,t}}{(1+r)^h}(1+\gamma)^h \theta_{h+a}}{\sum_{h=0}^{R-a} \frac{S_{h+a,t+h}/S_{a,t}}{(1+r)^h}(1+\gamma)^h \theta_{h+a}} \tau_{t+h} \tag{6b}$$

$$\mathscr{R}_{a,t} = \sum_{h=R+1}^{T} \frac{S_{h,t+h-a}/S_{a,t}}{(1+r)^{h-a}} p_{t+R+1-a,t+h-a}. \tag{6c}$$

Under CRRA preferences ($u = c^{1-\sigma}/(1-\sigma)$), the consumption profile of a generation born at t is proportional to wealth at birth—that is,

$$c_{h,t+h} = \Phi_{h,t}\left(\mathscr{W}^d_{0,t} + \mathscr{R}_{0,t}\right), \text{ with } \Phi_{h,t} \equiv \frac{(\beta(1+r))^{h/\sigma}}{\sum_{s=0}^{T} \frac{(\beta(1+r))^{s/\sigma}}{(1+r)^s} S_{s,t+s}}. \tag{7}$$

A special case occurs when $\beta(1+r) = 1$, which implies constant consumption:

$$c = \frac{1}{\sum_{s=0}^{T} \beta^s S_{s,t+s}}\left(\mathscr{W}^d_{0,t} + \mathscr{R}_{0,t}\right).$$

To analyze the generational distributional effects of a PAYG social security system, we rewrite equation (7) to express each generation's consumption profile as a function of its current wealth and the return on the remaining social security contributions. Generation a's wealth is composed of their current assets, the present value of their remaining labor income, and the present value of future pensions. That is, $(1+r)b_{a-1,t} + \mathscr{W}^d_{a,t} + \mathscr{R}_{a,t}$. Letting the gross return on pension contributions for the remainder of active life be

$$\hat{r}_{a,t} = \frac{\mathscr{R}_{a,t}}{\hat{\tau}_{at}\mathscr{W}_{at}} = \frac{\sum_{h=R+1}^{T} \frac{S_{h,t+h-a}}{(1+r)^{h-a}} p_{t+R+1-a,t+h-a}}{\sum_{h=a}^{R} \frac{S_{h,t+h-a}}{(1+r)^{h-a}} \tau_{t+h-a} W_{t+h-a} \theta_h}, \tag{8}$$

generation a's wealth can be expressed as $(1+r)b_{a-1,t} + \mathscr{W}_{a,t} + (\hat{r}_{a,t} - 1)\,\hat{\tau}_{at}\mathscr{W}_{at}$. Using this expression, their future consumption profile is

$$c_{a+h,t+h} = \Phi^a_{h,t}\left((1+r)b_{a-1,t} + \mathscr{W}_{a,t} + (\hat{r}_{a,t} - 1)\,\hat{\tau}_{at}\mathscr{W}_{at}\right) \tag{7'}$$
$$\text{with } \Phi^a_{h,t} \equiv \frac{(\beta(1+r))^{h/\sigma}}{\sum_{s=0}^{T-a}\frac{(\beta(1+r))^{s/\sigma}}{(1+r)^s}\frac{S_{a+s,t+s}}{S_{a,t}}} \text{ for } 0 \le h \le T-a.$$

An important implication of (7') is that the consumption path for the remaining lifetime of the generation aged a at t is a monotonically increasing function of the social security parameters through the term $(\hat{r}_{a,t} - 1)\,\hat{\tau}_{a,t}$.

That ratio—the present value of pension payments per unit of the present value of remaining contributions—is central to our analysis. To build intuition, suppose death is certain at 85, work starts at 25 and lasts 40 years, and $r = 0$. Agents contribute for 40 years and collect pensions for 20, so yearly pensions must be twice yearly contributions for social security to break even at birth. Mortality risk after 65—the triangle at the top of the population pyramid—cuts expected retirement years to roughly 10, doubling the required pension again. A positive interest rate, which discounts the late-in-life benefits most heavily, raises the break-even pension-to-contribution ratio further. Sections 3.2 and 4.2 make these calculations precise.

2.3 Firms

Firms choose labor L_t and capital K_t to maximize profits given wages, interest rates, and the linearly homogeneous neoclassical production function with labor-augmenting technical change $F(K_t, A_t L_t)$. Profits are $F(K_t, (1+\gamma)^t L_t) - W_t L_t - (r+\delta)K_t$. Writing the production function as $(1+\gamma)^t f(k_t) L_t$ for $k_t = \frac{K_t}{(1+\gamma)^t L_t}$, profits are $(1+\gamma)^t L_t\left(f(k_t) - \frac{W_t}{(1+\gamma)^t} - (r+\delta)k_t\right)$. Profit maximization requires

$$r + \delta = f'(k_t) \tag{9a}$$
$$W_t = (1+\gamma)^t\left(f(k_t) - f'(k_t)\,k_t\right). \tag{9b}$$

The first-order condition (9a) determines k_t given r, which is time invariant. As the labor supply is inelastic, the first-order condition for labor demand, (9b), determines equilibrium wages. For any $0 \le s \le t$, we can write $W_t = W_s(1+\gamma)^{t-s}$.

The solution to the firm's problem in an environment in which the interest rate is fixed by the small-open-economy assumption and labor is inelastically supplied implies that wages and output are essentially exogenous.

2.4 The Government: A Social Security System

The government in this economy is a pay-as-you-go social security transfer system (SS). It collects labor income taxes from workers and transfers resources to retirees. If the aggregate transfers in any period do not balance, the government issues risk-free debt, and it must satisfy an intertemporal budget constraint.

The SS has three possibly time-dependent policy parameters, $\Pi_t = \{P_t, \tau_t, R_t\}$: the replacement ratio, the labor income tax rate, and the retirement age, which are common across generations.[2] We model the SS pension received by the generation that retired at t as a linear function of their wage at their retirement date,

$$p_{t,t+h} = P_{t+h} W_t. \tag{10}$$

The aggregate SS outlays are the sum of each currently retired generation's pension benefits—that is, $\sum_{h=R+1}^{T} \mathscr{P}_{h,t} P_t W_{t-(h-R-1)}$—which we can express as

$$\sum_{h=R+1}^{T} \mathscr{P}_{h,t} P_t W_{t-h+R+1} = P_t \Gamma_t^{-1} D_t W_t, \text{ with } \Gamma_t^{-1} = \sum_{h=R+1}^{T} \pi_{h,t}^{D} \frac{1}{(1+\gamma)^{h-R-1}}. \tag{11}$$

Aggregate social security outlays are the product of the replacement ratio, wages, the aggregate dependent population, and the adjustment term Γ^{-1}, which is a weighted average of the wage growth between each generation's retirement date and period t. For positive values of the growth rate, Γ satisfies $0 < \Gamma^{-1} < 1$.

Aggregate SS revenues are the sum of the labor income taxes paid by all active workers $\sum_{h=0}^{R} \tau_t \theta_h W_t \mathscr{P}_{h,t}$, which we express as

$$\sum_{h=0}^{R} \tau_t \theta_h \mathscr{P}_{h,t} W_t = \tau_t N_t \bar{\theta}_t W_t, \text{ with } \bar{\theta}_t = \sum_{h=0}^{R} \pi_{ht}^{N} \theta_h. \tag{12}$$

The vector with elements θ_h describes the effective labor efficiency units supplied by each cohort aged h. Effective labor units vary with age throughout the life cycle.[3] The weighted average $\bar{\theta}_t$ depends on time through the age distribution of the active population. Notice that the effective labor force is $L_t = \bar{\theta}_t N_t$, so aggregate revenues in (12) boil down to the natural expression $\tau_t W_t L_t$.

[2] P_t can accommodate a range of institutional features, including reference-wage and indexation rules; see Appendix A.

[3] In some experiments, they depend on time because of changes in female labor force participation.

Combining the formulas for SS outlays and revenues, we can write the SS deficit as

$$d_t = P_t \Gamma_t^{-1} D_t W_t - \tau_t N_t \bar{\theta}_t W_t \tag{13a}$$

$$= \left(\frac{P_t}{\tau_t} \frac{\Gamma_t^{-1} D_t}{N_t \bar{\theta}_t} - 1 \right) \tau_t N_t \bar{\theta}_t W_t. \tag{13b}$$

The second expression for the deficit, (13b), shows that the total deficit is the product of the excess generosity ratio (deficit per dollar raised) and the scale of the system, as measured by its total revenues. The balanced-budget SS generosity ratio,

$$\frac{P_t}{\tau_t} = \frac{N_t \bar{\theta}_t}{\Gamma_t^{-1} D_t}, \tag{14}$$

equals the old-age support ratio N_t/D_t adjusted for the life-cycle labor-supply effect $\bar{\theta}_t$ and for wage growth Γ_t.

The government's budget constraint combines the difference equation that governs government debt with a boundary condition specifying the initial debt level and a boundary condition that prevents Ponzi schemes. For an arbitrary $\bar{B}_0$,

$$B_{t+1} = (1+r)(B_t + d_t) \text{ for } t = 0, 1, \ldots \tag{15a}$$

$$B_0 = \bar{B}_0 \tag{15b}$$

$$\lim_{t \to \infty} (1+r)^{-t} B_{t+1} \leq 0. \tag{15c}$$

Definition 1 (Fiscally feasible SS policy) *A fiscally feasible social security policy is a sequence* $\Pi = \{\Pi_t\}_{t \geq 0}$ *such that the government budget constraints* (15) *are satisfied.*

2.4.1 A Natural Borrowing Limit

The small-open-economy assumption and inelastic labor supply together imply that aggregate output is, from the government's fiscal perspective, effectively an endowment. Equation (9a) pins down the capital-labor ratio at $k^* = (f')^{-1}(r+\delta)$ independently of policy, and since the effective labor force $L_t = \bar{\theta}_t N_t$ is determined entirely by demographics, aggregate output satisfies $Y_t = (1+\gamma)^t f(k^*) L_t$ and evolves exogenously. Under a Cobb-Douglas production function with capital share α, the wage bill is a constant fraction of output: $W_t L_t = (1-\alpha) Y_t$.

Iterating (15a) forward from any date t and applying the no-Ponzi condition (15c) with equality yields $B_t = \sum_{s=t}^{\infty} (1+r)^{-(s-t)} s_s$, where $s_s = -d_s$ is the primary surplus at s. Since pension outlays are non-negative and the tax rate is at most one, the surplus satisfies $s_t \leq \tau_t W_t L_t \leq W_t L_t = (1-\alpha) Y_t$. The upper bound is achieved only in the edge case in which pensions are fully eliminated

($P_t = 0$) and all labor income is taxed ($\tau_t = 1$). This case represents the maximum resources the government could ever redirect to debt service. For this bound we approximate the effective labor force as growing at a constant rate n around its demographic trend so that $Y_s/Y_t \approx [(1+\gamma)(1+n)]^{s-t}$. Bounding the surplus at $(1-\alpha)Y_t$ and summing yields the *natural borrowing limit*, which must hold at every date t:

$$B_t \leq \sum_{s=t}^{\infty} \frac{(1-\alpha)Y_s}{(1+r)^{s-t}} = \frac{(1-\alpha)(1+r)}{r-\gamma-n} Y_t \equiv \bar{B}_t, \tag{16}$$

where convergence requires $r > \gamma + n$—the condition for dynamic efficiency, maintained throughout. As a ratio to contemporaneous GDP, the limit is constant:

$$\frac{\bar{B}_t}{Y_t} = \frac{(1-\alpha)(1+r)}{r-\gamma-n}. \tag{17}$$

The no-Ponzi condition (15c), read as a borrowing constraint that rules out unbounded debt growth, is equivalent to the natural borrowing limit holding at every period.[4]

2.5 Definition of Equilibrium

Definition 2 (Competitive equilibrium) *Given an initial age distribution of the population π_0, a sequence of current and future age-dependent fertility rates and survival probabilities $\{\phi_t, S_t\}_{t\geq 0}$, the interest rate r, the growth rate γ, the labor efficiency units θ, and the fiscally feasible social security policy Π, a competitive equilibrium allocation is a sequence of wages satisfying (9) and a sequence of consumption allocations $\{c_{a,t}\}_{a=0}^{T}$ for all $t \geq 0$ given by (7').*

The definition of equilibrium does not include an aggregate consistency condition because, in the small open economy, aggregate consistency follows from the fact that households and the government satisfy their budget constraints at the world interest rate. Aggregate consistency implies the accumulation of net foreign assets.

3. The Distributional Impact of Social Security

Consider policies $\Pi_t = \{P_s, \tau_s, R_s\}_{s\geq t}$ and $\Pi'_t = \{P'_s, \tau'_s, R'_s\}_{s\geq t}$. Let $V_{a,t}(c)$ denote generation a's lifetime utility at date t from consumption path $c = \{c_{a+h,t+h}\}_{h=0}^{T-a}$, given by the left-hand side of equation (2), and let $c_a(\Pi_t)$ denote generation a's equilibrium consumption path under policy

[4] Iterating (15a) forward from t and bounding the surplus at $(1-\alpha)Y_t$ gives $B_t \leq \bar{B}_t$ for all t; conversely, $B_t \leq \bar{B}_t$ implies $(1+r)^{-t}B_t \leq (1+r)^{-t}\bar{B}_t = (1-\alpha)(1+r)/(r-\gamma-n)\cdot Y_0 \cdot ((1+\gamma)(1+n)/(1+r))^t \to 0$ as $t \to \infty$, which is precisely (15c).

Π_t. We write $V_{a,t}(\Pi_t) \equiv V_{a,t}(c_a(\Pi_t))$ for generation a's equilibrium utility under policy Π_t. We examine $\Delta V_{a,t} = V_{a,t}(\Pi'_t) - V_{a,t}(\Pi_t)$, the change in welfare from switching from policy Π to policy Π' for generation a at time t.

First, we develop a formula for the equilibrium percentage change in permanent consumption from switching from policy Π to Π' for each generation, denoted by $\lambda_{a,t}$, defined by

$$V_{a,t}\left(c_a(\Pi'_t)\right) = V_{a,t}\left(c_a(\Pi_t)(1+\lambda_{a,t})\right).$$

$\lambda_{a,t}$ is the analog of Lucas's welfare-equivalent measure for evaluating policy: the consumption-equivalent gain from switching to Π', or the permanent change in consumption after t that makes generation a indifferent between the two policies.

The equilibrium consumption profile following date t for generation a is proportional to their wealth at t as shown in (7'), which we reproduce here for convenience: $c_{a+h,t+h} = \Phi^a_{h,t}\left((1+r)b_{a-1,t} + \mathscr{W}_{a,t} + (\hat{r}_{a,t}-1)\,\hat{\tau}_{a,t}\mathscr{W}_{a,t}\right)$. After retirement, for $a > R$, the formula works under the convention that $\mathscr{W}_{a,t} = \hat{\tau}_{a,t} = 1$, so that $c_{a+h,t+h} = \Phi^a_{h,t}\left((1+r)b_{a-1,t} + \mathscr{R}_{a,t}\right)$, with $\mathscr{R}_{a,t}$ redefined as the present value of the remaining pension benefits.

Let $c_{a+h,t+h}$ and $c'_{a+h,t+h}$ be equilibrium consumption under policies Π and Π', respectively. Then, equation (7') implies that for any future period $0 \le h \le T-a$, the percentage change in consumption under the two policies is

$$\lambda_{a,t} \equiv \frac{c'_{a+h,t+h} - c_{a+h,t+h}}{c_{a+h,t+h}} = \frac{(1+r)b_{a-1,t} + \left[1+\hat{\tau}'_{a,t}\left(\hat{r}'_{a,t}-1\right)\right]\mathscr{W}_{a,t}}{(1+r)b_{a-1,t} + \left[1+\hat{\tau}_{a,t}\left(\hat{r}_{a,t}-1\right)\right]\mathscr{W}_{a,t}} - 1 \tag{18}$$

with the appropriate modification for retirees—that is, $a > R$. Notice that $\lambda_{a,t}$ is independent of h and thus permanent.

Consider policies that keep the retirement age constant.[5] For working-age individuals, equation (18) reduces to

$$\lambda_{a,t} = \frac{\left(\hat{r}'_{a,t}-1\right)\hat{\tau}'_{a,t} - \left(\hat{r}_{a,t}-1\right)\hat{\tau}_{a,t}}{\frac{(1+r)b_{a-1,t}}{\mathscr{W}_{a,t}} + 1 + \hat{\tau}_{a,t}\left(\hat{r}_{a,t}-1\right)}. \tag{19}$$

The welfare effect of a SS policy change for generation a depends on the change in the product of their remaining average contribution per dollar earned, $\hat{\tau}_{a,t}$, and the net return on those contributions, $\hat{r}_{a,t} - 1$.

[5] Policies that change the retirement age R have two opposing effects. They reduce the return on remaining contributions, $\hat{r}_{a,t}$, by reducing the number of periods accruing benefits and increasing the number of contribution periods. At the same time, they increase the present value of wages. Our model is ill suited to evaluate the benefits of increased labor income because it lacks a value for leisure and does not account for retirement decisions. Analyzing a model with labor-leisure choices remains for future research.

3.1 Comparing Social Security to Laissez-Faire

We now apply this framework to evaluate the welfare effects of SS relative to laissez-faire. Consider the SS policy $\Pi_t = \{P_s, \tau_s, R\}_{s \geq t}$ and the laissez-faire saving environment $\Pi_t^{LF} = \{0, 0, R\}_{s \geq t}$. The welfare gain from replacing SS with laissez-faire for generation a at time t—the percentage change in lifetime consumption—is equal to the net loss on a's remaining SS contributions, $\hat{\tau}_{a,t}\mathscr{W}_{a,t}$, as a share of their wealth under the status quo, $(1+r)b_{a-1,t} + [1 + \hat{\tau}_{a,t}(\hat{r}_{a,t} - 1)]\mathscr{W}_{a,t}$:

$$\lambda_{a,t} = \hat{\tau}_{a,t}(1 - \hat{r}_{a,t}) \frac{\mathscr{W}_{a,t}}{(1+r)b_{a-1,t} + [1 + \hat{\tau}_{a,t}(\hat{r}_{a,t} - 1)]\mathscr{W}_{a,t}}. \tag{20}$$

It follows that $\hat{r}_{a,t} - 1$ is a sufficient statistic to determine which generations win and which ones lose from SS.

Lemma 1 (Sufficient statistic) *The return on pension contributions for the remainder of active life $\hat{r}_{a,t}$ is a sufficient statistic to determine the winners and losers from SS.*

This sufficient-statistic property is specific to the comparison against laissez-faire. It does not extend to comparisons between two reforms, where $\lambda_{a,t}$ depends on the full product $\hat{\tau}_{a,t}(\hat{r}_{a,t} - 1)$ under each policy (equation (19)), so two reforms need not be ranked the same way by $\hat{r}$ and by λ.

Introducing the policy parameters in the definition of $\hat{r}_{a,t}$ yields

$$\hat{r}_{a,t} = \left(\frac{1+\gamma}{1+r}\right)^{R+1-a} \frac{\sum_{h=0}^{T-R-1} \frac{S_{R+1+h,t+R+1-a+h}}{(1+r)^h} P_{t+R+1-a+h}}{\sum_{h=0}^{R-a} \frac{S_{a+h,t+h}}{(1+r)^h} \tau_{t+h}(1+\gamma)^h \theta_{a+h}}. \tag{21}$$

Computing the generational return to SS requires knowledge of survival probabilities for the remainder of life, SS tax rates until retirement (between a and R), and the replacement ratios in place after $R+1$.

It will be useful to understand the properties of SS's return $\hat{r}_{a,t}$ as a function of age. As agents age, the present value of their pension benefits is higher because it is discounted over fewer periods, while the present value of their contributions falls as they contribute for fewer years. As a result, the ratio of the present value of benefits to the present value of contributions rises with age. In a stationary environment, the following lemma follows directly from equation (21).

Lemma 2 (Social Security favors the old) *For $a < R$, given parameters r, γ, $\{P_{t+R+1-a+h}\}_{h=0}^{T-R-1}$, and $\{\tau_{t+h}\}_{h=0}^{R-a}$, the return on SS at time t for agents aged a is a monotonically increasing function of a, reaching its maximum at $a = R$.*

In the non-stationary environments of Sections 4.3 and 4.4, we verify numerically that the same monotonicity holds. We define the break-even generation as the generation indifferent between SS and laissez-faire.

Definition 3 (Break-even generation) *Consider an SS system for which the return at birth satisfies $\hat{r}_{0,t} \leq 1$. The break-even generation, $\bar{a}_t$, is the generation indifferent between the two policies:*

$$\hat{r}_{\bar{a}_t,t} = 1 \iff V_{\bar{a}_t,t}(\Pi_t) = V_{\bar{a}_t,t}(\Pi_t^{LF}). \tag{22}$$

The existence of the break-even generation is guaranteed by Lemma 2. If the solution to equation (22) is not an integer, $\bar{a}_t = \min\{a : \hat{r}_{a,t} \geq 1\}$.

3.2 Balanced-Budget Steady-State Welfare Analysis

This section considers a steady state in which the population's age structure is stationary and the SS budget is balanced. The generosity ratio in (14) becomes

$$\frac{P}{\tau} = \frac{N\bar{\theta}}{D\Gamma^{-1}}. \tag{23}$$

The individual return on SS in (21) evaluated at the stationary balanced-budget generosity ratio is

$$\hat{r}_a = \left(\frac{1+\gamma}{1+r}\right)^{R+1-a} \frac{N\bar{\theta}}{D\Gamma^{-1}} \frac{\sum_{h=0}^{T-R-1} \frac{S_{R+1+h}}{(1+r)^h}}{\sum_{h=0}^{R-a} \frac{S_{a+h}}{(1+r)^h}(1+\gamma)^h \theta_{a+h}}. \tag{24}$$

Workers contribute to the system an amount proportional to the effective labor force $(N_t\bar{\theta})$ for $R+1$ periods, and retirees collect pensions proportional to the retired population $(D_t\Gamma^{-1})$ for $T-R$ periods. The streams of benefits and contributions are discounted at the age-dependent discount factors $S_h(1+r)^{-h}$.

We can collect the demographic variables in (24) in the terms

$$\mathscr{S} = \sum_{h=0}^{T-R-1} \frac{S_{R+1+h}}{(1+r)^h} \tag{25a}$$

$$\mathscr{L}_a = \sum_{h=0}^{R-a} \frac{S_{a+h}}{(1+r)^h}(1+\gamma)^h \theta_{a+h} \tag{25b}$$

$$(1+\tilde{n}_a)^{R+1-a} = \frac{N\bar{\theta}}{D\Gamma^{-1}} \frac{\mathscr{S}}{\mathscr{L}_a} = \frac{P}{\tau} \frac{\mathscr{S}}{\mathscr{L}_a}, \tag{25c}$$

where $\mathscr{S}$ is an operator to compute the expected present value of retirement years and $\mathscr{L}$ is an operator to compute the expected present value of the wage stream between age a and R. Using this definition, we can rewrite equation (24) as

$$\hat{r}_a = \left(\frac{(1+\gamma)(1+\tilde{n}_a)}{1+r} \right)^{R+1-a}, \tag{24'}$$

which resembles the textbook expression in Blanchard and Fischer (1989). Observe that $(1+\tilde{n}_a)^{R+1-a}$ is the product of the steady-state generosity ratio and the ratio of the expected present value of a unit of pension benefits at the retirement age to the expected present value of contributions at age a, so

$$\hat{r}_a = \left(\frac{1+\gamma}{1+r} \right)^{R+1-a} \frac{\mathscr{S}}{\mathscr{L}_a} \frac{P}{\tau}. \tag{24''}$$

The product $P\mathscr{S}$ is the expected present value of the wage replacement for the whole retirement period, $(1+\gamma)^{R+1-a}$ captures the wage at retirement, and $(1+r)^{-(R+1-a)}$ discounts it. The product $\tau\mathscr{L}$ is the expected present value of SS contributions over the whole contributing period, taking into account labor productivity growth (γ) and efficiency units of labor force participation (θ).

3.3 Social Security as a Zero-Sum Game

The welfare formula (19) evaluates a policy one cohort at a time. Adding across cohorts gives three results that organize the quantitative analysis. The net social security wealth of all cohorts, present and future, sums to the present value of the system's deficits (Lemma 3). Under the maintained assumption $r > \gamma + n$, the competitive equilibrium under any fiscally feasible policy is Pareto optimal, so no reform can make every cohort better off (Proposition 1). And between two fiscally feasible policies with the same retirement age, a reform that a majority prefers lowers the aggregate wealth of the outvoted minority and of future cohorts by exactly the majority's gain, although how that loss is split between the two is unrestricted (Proposition 2 and Corollary 1).

As in Section 4.1, age 0 is labor market entry, so ages R and T are measured in years since entry and biological ages are obtained by adding 18. We index cohorts by the date s at which they enter economic life: Cohorts alive at reform date $t = 0$ have $s \in \{-T, \ldots, 0\}$ and age $a = -s$; *future cohorts* enter at $s \geq 1$ and include those younger than 18 at the reform date, who neither work, consume, nor vote. The *social security wealth* of the cohort aged a at t is the present value of its remaining benefits net of its remaining contributions,

$$\Psi_{a,t} \equiv \mathscr{R}_{a,t} - \hat{\tau}_{a,t}\mathscr{W}_{a,t} = (\hat{r}_{a,t} - 1)\,\hat{\tau}_{a,t}\mathscr{W}_{a,t}, \tag{26}$$

with $\Psi_{a,t} = \mathscr{R}_{a,t}$ and $\mathscr{W}_{a,t} = 0$ for retirees, and its *total wealth* is $\Omega_{a,t} \equiv (1+r)b_{a-1,t} + \mathscr{W}_{a,t} + \Psi_{a,t}$. By (7') the cohort's remaining consumption path is $\Phi^{a}_{h,t}\Omega_{a,t}$, and (18) reduces to $1+\lambda_{a,t} = \Omega_{a,t}(\Pi')/\Omega_{a,t}(\Pi)$. A reform announced at t leaves $b_{a-1,t}$ and $\mathscr{W}_{a,t}$ unchanged, so a cohort prefers it if and only if it raises $\Psi_{a,t}$. Write $\Psi_s(\Pi)$ for the social security wealth of cohort s: $\Psi_{-s,0}(\Pi)$ for those alive ($s \le 0$) and $\Psi_{0,s}(\Pi)$ for the unborn ($s \ge 1$). Weight each cohort by its size at the reform date, discounted for future cohorts,

$$\nu_s \equiv \begin{cases} \mathscr{P}_{-s,0}, & s \le 0, \\ (1+r)^{-s}\,\mathscr{P}_{0,s}, & s \ge 1, \end{cases} \tag{27}$$

and call $\Psi(\Pi) \equiv \sum_{s \ge -T} \nu_s \Psi_s(\Pi)$ the *aggregate social security wealth* of policy Π.

Two conditions are maintained throughout this subsection. First, the survival probabilities that price survival-contingent claims in (4) also count survivors: Of the cohort aged a at t,

$$\mathscr{P}_{a+h,t+h} = \frac{S_{a+h,t+h}}{S_{a,t}}\,\mathscr{P}_{a,t} \qquad \text{members are alive at } t+h. \tag{28}$$

Second, present values converge:

$$\sum_{t \ge 0} (1+r)^{-t} W_t\, \mathscr{P}_t < \infty. \tag{29}$$

This is the condition $1+r > (1+\gamma)(1+n)$, or $r > \gamma + n + \gamma n$—the exact form of the approximate inequality $r > \gamma + n$ maintained in Section 2.4.1—applied to total population rather than to effective labor; the two grow at the same rate in the long run when age-specific survival is fixed. Because $0 \le \tau_t \le 1$, $L_t \le (\max_h \theta_h)\,\mathscr{P}_t$, and pensions are bounded relative to wages, (29) makes the present values of the wage bill, contributions, and pensions absolutely convergent.

Lemma 3 (Zero-sum identity) *Under* (28) *and* (29)*, for any policy* Π*,* $\Psi(\Pi) = \sum_{t \ge 0}(1+r)^{-t} d_t(\Pi)$*. Hence* Π *is fiscally feasible if and only if* $\Psi(\Pi) = -\bar{B}_0$*.*

SS is a closed system in present value: What all cohorts, present and future, expect to take out of it net of what they put in equals the present value of what it borrows. Starting from $\bar{B}_0 = 0$, as in Section 4.3, the gains of the winners are exactly paid for by the losses of the losers. The proof (Appendix E) does no more than assign every peso of every deficit to the cohort that receives or pays it.

Proposition 1 (Pareto optimality) *Under* (28) *and* (29)*, let* Π *be fiscally feasible. The competitive equilibrium allocation under* Π *is Pareto optimal among the allocations the economy can*

afford at the world interest rate: No consumption allocation that satisfies the economy's intertemporal resource constraint, equation (45) in Appendix E, gives every cohort, present and future, at least its equilibrium utility and some cohort strictly more.

The argument is the one Balasko and Shell (1980) give, in the remark following their Proposition 5.3, for a pure-exchange overlapping-generations economy: A competitive allocation whose value at the supporting prices is finite is Pareto optimal. Our environment adds production, foreign borrowing, survival-contingent claims, and government transfers, so Appendix E first collapses it to a single resource constraint at the supporting prices, with a finite right-hand side equal to national wealth plus the present value of the wage bill, then runs their argument.

The proof uses only strictly increasing utility and the finiteness of aggregate wealth, (29). When aggregate wealth is infinite, the argument proves nothing, and this is the territory in which a PAYG transfer can be a Pareto improvement over laissez-faire (Samuelson, 1958; Aaron, 1966). Condition (29) is sufficient; we make no claim about the boundary case $1+r=(1+\gamma)(1+n)$.

The economic environment in this paper is, essentially, an endowment economy, in which policy reallocates resources across generations without destroying any. In an economy with elastic labor supply a payroll tax distorts, and a phase-out can generate a surplus to distribute—the channel described in Section 6(i).

Proposition 1 implies that no fiscally feasible transition to laissez-faire or a funded system can be Pareto improving (Breyer, 1989). Among fiscally feasible policies with the same retirement age, the choice of policy is therefore a pure division of a fixed aggregate across cohorts. Two of the comparisons in Section 4 fall outside this statement: The business-as-usual benchmark is not fiscally feasible—it crosses the natural borrowing limit—so a welfare change measured relative to it compares a feasible policy with one that cannot be continued, which is why fiscal adjustment can lower every cohort's welfare; and a change in the retirement age changes the labor endowment, and thus the aggregate being divided, so Proposition 1 applies to each retirement age separately.

Proposition 2 (Majority reforms are transfers) *Let Π and Π' be fiscally feasible with the same $\bar{B}_0$ and the same retirement age, and let $\Delta\Psi_s \equiv \Psi_s(\Pi') - \Psi_s(\Pi)$. Then $\sum_s \nu_s \Delta\Psi_s = 0$. Consequently, for any set of cohorts M, if $\Delta\Psi_s \geq 0$ for all $s \in M$ with strict inequality for some $s \in M$, then*

$$\sum_{s \notin M} \nu_s \Delta\Psi_s \;=\; -\sum_{s \in M} \nu_s \Delta\Psi_s \;<\; 0. \tag{30}$$

Apply the proposition to a reform adopted by majority rule. Let M be the set of living cohorts that vote for it—a majority of voters, none of whom belong to a future cohort—so that its complement consists of the outvoted living cohorts and of all future cohorts.

Corollary 1 (Who pays for a majority's reform) *(a) Any fiscally feasible reform that makes every cohort in M weakly better off, and some strictly better off, makes at least one cohort outside M strictly worse off: The complement's aggregate social security wealth falls by exactly the majority's aggregate gain. (b) Either part of the complement can pay. Suppose, departing from Section 2.4, that contribution rates and replacement ratios may differ across cohorts; that under Π every cohort's remaining contribution rate lies in $[\underline{\tau},\overline{\tau}]$ with $0<\underline{\tau}\leq\overline{\tau}<1$, every working-age cohort, present or future, has positive remaining wage wealth, and every retiree has $\mathscr{R}_{a,0}>0$; and that the set of outvoted living cohorts is non-empty. Then for every sufficiently small $\delta>0$, each of the following is fiscally feasible with contribution rates in $[0,1]$ and non-negative pensions: (i) every cohort in M gains δ, the outvoted living cohorts lose, and future cohorts are unaffected; (ii) every cohort in M gains δ, the outvoted living cohorts are unaffected, and future cohorts lose; (iii) every cohort in M gains δ, the outvoted living cohorts lose more than the majority gains, and future cohorts gain. The only excluded pattern is one in which every cohort outside M is weakly better off.*

Part (a) restates Proposition 2; part (b) is proved in Appendix E by cohort-specific changes in contributions or benefits and is a statement about an instrument set richer than this paper's. The size of the majority's gain is immaterial: A small gain requires a correspondingly small aggregate loss elsewhere, but its sign is forced.

Our policy parameters are common across cohorts, which restricts who can be made to pay. A cohort entering at $s\geq 1$ receives its pension from date $s+R+1$ on, while every cohort alive at the reform date has retired by date $R+1$: Changes in P_t confined to $t\leq R+1$ therefore touch only the living, and changes in τ_t confined to $t\geq R+1$ fall only on future cohorts. The sign pattern of (ii) is thus available with common instruments. Raise P_t by ΔP for $0\leq t\leq k$, with $k\leq R+1$, benefiting exactly the cohorts within k years of retirement; leave τ_t unchanged through date R, so the outvoted young are exactly unaffected; and raise it from date $R+1$ on by the constant

$$\Delta\tau=\Delta P\,\frac{\sum_{t=0}^{k}(1+r)^{-t}\,\Gamma_t^{-1}D_tW_t}{\sum_{t\geq R+1}(1+r)^{-t}\,W_tL_t}, \tag{31}$$

which satisfies (15) exactly by Lemma 3. If ΔP is small enough, $\Delta\tau$ stays below $1-\overline{\tau}$, and if the status quo's debt-to-GDP ratio is bounded away from the natural borrowing limit (16), the reformed path respects it too. With a last working age of 65 and benefits paid from 66, $k=25$ makes the gainers the cohorts aged 41 and above—including Argentina's 2023 median voter (Section 5)—while the entire bill falls on cohorts that enter the labor force after every voter alive today has retired: the compulsory Ponzi scheme in its purest form. Pattern (i), which spares future cohorts, instead requires financing the majority's gain with a levy at the reform date on every living worker; with cohort-specific instruments, Corollary 1(b) holds for any age distribution.

4. Quantitative Implications

This section quantifies the implications of current demographic trends for the generational distribution and sustainability of the Argentine social security system.

4.1 Data, Calibration, and Simulation Setup

What is a reasonable growth rate? We estimate γ from the regression $\ln y_t^i = c + \frac{\gamma}{100}t + \varepsilon_t$ on Argentine and US output per capita and per worker over several sample periods (Table 1). The US growth rate exceeds Argentina's in every sample and source.[6] As the US is the world's industrial leader, we take its growth rate as Argentina's potential, which amounts to assuming that Argentina remains at its current income relative to the US.[7] For samples starting in 1950 the US estimate is close to 2% per year. Output per worker and per hour grew more slowly because the employment-to-population ratio rose by about a third between 1960 and 1990 with the rise in female labor force participation.[8] Longer samples from Maddison's historical statistics (Bolt and van Zanden, 2024) also give less than 2%, as do Fernández-Villaverde et al. (2025) for 1991–2019.

Table 1: Long-Term Growth Rate

Sample period	γ^{US}	γ^{ARG}	Source
1875-2022	1.68	1.21	Maddison - output per capita
	(0.02)	(0.02)	
1950-2022	1.92	1.20	Maddison - output per capita
	(0.03)	(0.06)	
1950-2023	1.97	0.90	PWT - output per capita
	(0.03)	(0.05)	
1950-2023	1.63	0.60	PWT - output per worker
	(0.02)	(0.06)	
1950-2023	1.81	0.95	PWT - output per hour
	(0.02)	(0.05)	

Note: In each case, γ is the OLS estimate of the regression $\ln y_t = c + \frac{\gamma}{100}t + \varepsilon_t$, where y_t is output per capita, per worker, or per hour as indicated in the Source column. The variables γ^{US} and γ^{ARG} are estimated on the same sample period and output measure for the United States and Argentina, respectively. Standard errors in parentheses.
Sources: Maddison Project Database 2023, Bolt and van Zanden (2024), and Penn World Table, version 11.0 (Feenstra et al. (2015))

[6] The coefficient of the regression of $\log\left(y_t^{US}/y_t^{AR}\right)$ on a constant and time has a statistically significant positive time trend.

[7] This is the approach taken in all the country studies in Kehoe and Nicolini (2021).

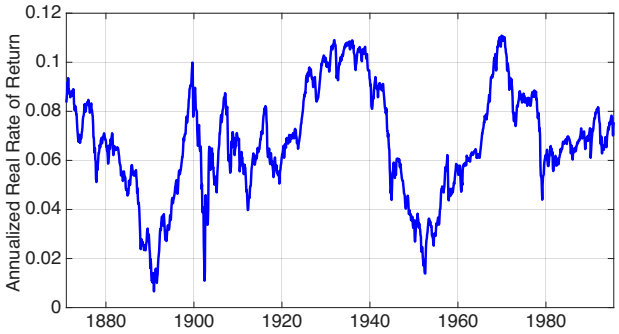
[8] See https://fred.stlouisfed.org/graph/?g=1Nphx for the employment-to-population ratio.

In our policy simulations we set $\gamma = 2\%$ for both the post-1950 period and the future. The estimate is generous and gives the SS scheme its best chance.

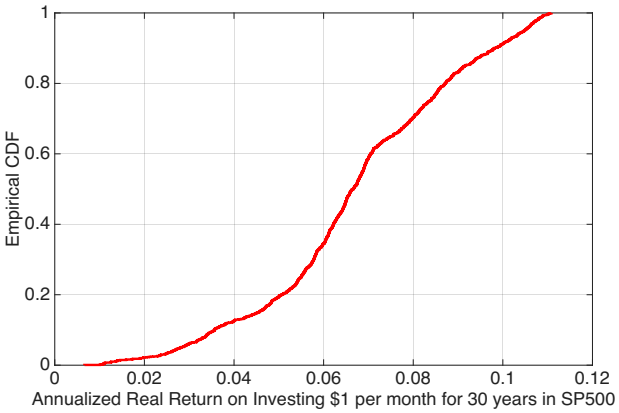
What is a reasonable interest rate? For a small open economy, the natural opportunity cost of contributing to a PAYG system is the return on global capital markets. We consider investing a constant real amount each month for 30 years in the S&P 500, reinvesting dividends, and compute the realized internal rate of return of every such 30-year window in Robert Shiller's (2025) dataset, with start dates from January 1871 to April 1995. Figure 1 plots these returns and their empirical CDF: The mean and median are 6.7% per year, with a standard deviation of 2.2%.

Figure 1: Implicit 30-Year Stock Market Return

(a) Internal Rate of Return on Investing Monthly in the S&P 500 for 30 Years, Starting at t



(b) Cumulative Distribution of Returns Shown in Panel (a).



$1 + r_{t+1} = \frac{p_{t+1} + d_{t+1}}{p_t}$
Source: Shiller (2025)

Table 2 locates candidate benchmarks in this distribution: 2% per year sits at the 2.2nd percentile of realizations and 8% at the 70th. We therefore work with a very conservative 4% and a moderately conservative 6%, the 13th and 35th percentiles.

Table 2: Frequency of Benchmark Returns

% return	**2**	**4**	**6**	**8**
Percentile	2.2	12.6	34.5	70.3

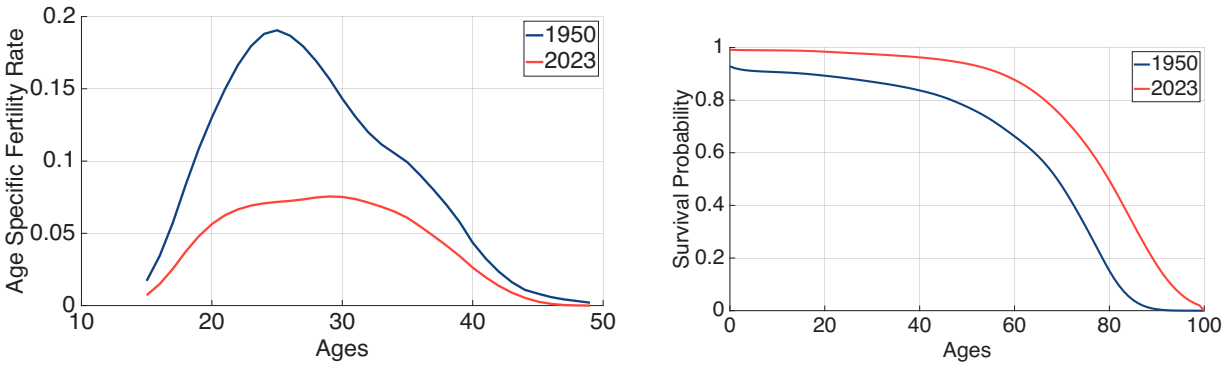
Note: Selected observations from Figure 1b.

Risk adjustment does not overturn these choices: Computing the certainty equivalent of the wealth accumulated by the strategy under CRRA preferences (Appendix D), our benchmark 6% is the certainty-equivalent return of a household with risk aversion $\sigma = 2.7$, while the conservative

4% corresponds to $\sigma = 8.4$; at the calibrated $\sigma = 2$, the certainty-equivalent return is 6.3%.[9]

Demography. We use historical demographic data from the United Nations *World Population Prospects* 2024, covering 1950–2023, and discard the projections, which extend from 2024 through 2100.[10]

Figure 2: Fertility and Survival in Argentina, 1950 and 2023



(a) Age-Dependent Fertility Rates **(b) Age-Dependent Survival probabilities**

Note: Fertility and survival profiles shown are from the data (not simulated). *Source:* United Nations, Department of Economic and Social Affairs, Population Division (2024)

Figure 2 displays the primitives that shape population dynamics: age-dependent fertility rates $\phi_{h,t}$ and unconditional survival probabilities S_{ht} for the years 1950 and 2023. Panel 2a shows how the average number of children per woman during her fertile years has plummeted between 1950 and 2023. Survival probabilities, on the other hand, have increased for all ages as shown in Panel 2b. For these primitives, simulated data from the demographic model in (1) replicate the total fertility rate[11] and natural population growth rates well.

[9] As a further reference, we estimate the US marginal product of capital at about 9% per year from US national accounts (1929–2025) and 8% per year from the Penn World Table (1950–2023); with a capital share of $1/3$ instead of 0.4, the net marginal product falls to 7.5% per year. National Income and Product Accounts' marginal product of capital is computed as $r = \alpha Y/K - \delta$, where Y is nominal GDP, K is the capital stock, $\alpha = 0.4$ is the capital share of income, and δ is the depreciation rate implicit in the Bureau of Economic Analysis's accounts; the first term is the marginal product of capital under a Cobb-Douglas production function. The Penn World Table's marginal product of capital is the series "irr" in the Penn World Table version 11.0, computed as $r = \frac{\text{Capital Income}}{K} - \delta$. Sources: US Bureau of Economic Analysis, Gross Domestic Product; K is the net stock of fixed assets and $\delta = D/K$, where D is the current-cost depreciation of fixed assets (GDPA, K1TTOTL1ES000 and M1TTOTL1ES000, retrieved from FRED, Federal Reserve Bank of St. Louis). The Bureau of Economic Analysis and the Penn World Table compute K with the perpetual inventory method.

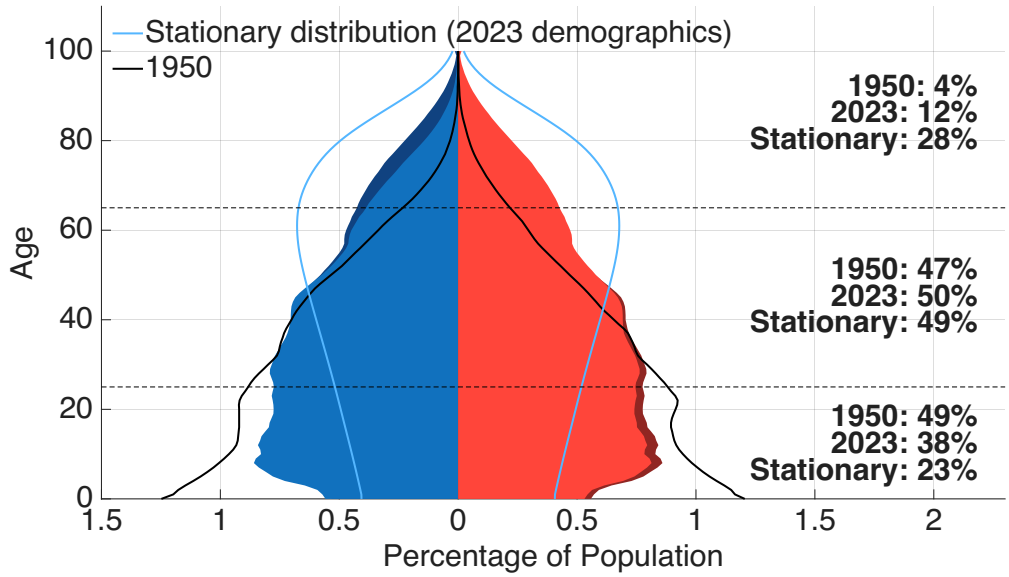
[10] National vital statistics and the UN WPP agree closely for Argentina. Onofri et al. (2026) compare the two birth series for the Latin American countries where both exist and find them nearly indistinguishable here (their Appendix Figure B.1).

[11] $TFR_t = \sum_{h \in \mathrm{R}} \phi_{ht}$, where R is the female reproductive period, which typically spans from 15 to 49 years of age, and ϕ_{ht} is the age-specific fertility rate.

The age distribution of the population, π_t, is represented in Figure 3's population pyramids for the years 1950 and 2023 and for the demographic steady state (circa 2100). The 2023 pyramid, computed by taking π_{1950} as an initial condition and iterating equation (1), approximates the empirical π_{2023}, with some differences attributed to migration flows. For the steady state, we compute π_∞ by simulating equation (1) forward from π_{2023}, holding the age-dependent fertility rates and survival probabilities at the 2023 levels depicted in Figure 2.[12]

Our computational experiments consider two steady-state distributions, which we label as fixed age distribution and fixed fertility. Fixed age distribution corresponds to the 2023 population pyramid, π_{2023}, and fixed fertility to π_∞ computed with the procedure detailed above. Both are depicted in Figure 3.

Figure 3: Argentina: Population Distribution, 1950, 2023, and Stationary Projection.



Note: The figure shows the Argentine population distribution by age and sex. The blue and red shaded areas represent, respectively, the male and female populations in 2023, as percentages of total population by single year of age (from 0 to 100). The black line displays the corresponding population distribution in 1950. The light blue line indicates the stationary age distribution implied by 2023 demographic parameters.
Source: United Nations, Department of Economic and Social Affairs, Population Division (2024)

The aging of the population that motivates this paper is reflected in the evolution of the population pyramids and plays a key role in the determination of the balanced-budget generosity ratio (equation (14)) and the PAYG's present value benefit per peso in equation (24'). While the

[12] When computing these demographic transitions, we assume that agents aged 0–18 do not work, consume, or vote, even though they are included in the population count and reproduction.

working-age population is roughly similar in the three cases, the share of the population over 65 tripled in the last 75 years and, ceteris paribus, will more than double for the next generation. The old-age dependency ratio increased from less than 0.1 in 1950 to 0.24 in 2023, and it is projected to reach 0.57.[13]

Preferences. Preferences are $u(c) = c^{1-\sigma}/(1-\sigma)$. We set $\sigma = 2$ and $\beta = 1/(1+r)$ at the benchmark $r = 6\%$, so $\beta(1+r) = 1$ and consumption remains flat over the life cycle in the absence of SS. In the exercises in which r varies, β is held fixed at its benchmark value. Consumption-equivalent welfare changes ($\lambda_{a,t}$) are reported only at $r = 6\%$.

Life-Cycle Efficiency Units. The construction of age-specific labor efficiency units follows Lagakos et al. (2018). Rather than estimating this profile ourselves, we import it directly from their preferred Heckman-Lochner-Taber decomposition (their Figure 5, right panel, and Table 4, Panel A), which uses cross-country survey data for a group of developing economies to separate the pure return to labor market experience from cohort and aggregate wage-growth effects—exactly the age component our model calls for since $w_{a,t} = W_t \theta_a$ has no cohort term and a single time effect W_t growing at γ. We treat their experience profile as our age profile θ_a, noting that in our model, age maps one-to-one onto labor market experience since all individuals enter the labor force at the same age and there is no unemployment, whereas this mapping does not hold exactly in the survey data underlying their estimates. Throughout this paper, model age $a = 0$ denotes labor market entry rather than biological birth: Economic life (supplying labor, paying contributions, and holding assets) begins at age 18, so model age a corresponds to biological age $a + 18$. We retain the terms *birth* and *newborn* for $a = 0$ as a modeling convention; $\hat{r}_{0,t}$ therefore denotes the implicit return to an 18-year-old entering the labor force, and reported ages such as the break-even age are biological ages ($a + 18$).

Current Policy. We calibrate the SS parameters (P, τ, R) to 2023 data under two alternative assumptions regarding fiscal balance. In the first, we take observed revenues and outlays at face value: SIPA contribution revenues amounted to 4.3% of GDP, while contributory pension outlays reached 5.6% of GDP (Subsecretaría de Seguridad Social, 2024), implying a deficit of approximately 1.3% of GDP financed through earmarked tax transfers and residual Treasury support.[14]

[13] This is the conventional demographic old-age dependency ratio, computed over the full empirical age distribution.

[14] The gap between contribution revenues and pension outlays is a structural feature of the Argentine system. Approximately 40% of workers are informal and contribute nothing (41% in the 2025 Encuesta Permanente de Hogares; Appendix B); employee contributions face a ceiling on taxable wages; and the self-employed under the *monotributo* regime pay a flat fee rather than a percentage of earnings. These factors compress effective contribution revenues well below what the statutory rate of 21%–23% applied to the full labor share would imply. As an independent cross-check on the outlay figure, approximately 5.7 million SIPA beneficiaries (ANSES, 2023) receive an average monthly benefit

In the second, we invoke the fungibility of revenue sources—earmarked payroll taxes, VAT co-participation transfers, and residual Treasury contributions all flow into the same National Social Security Administration budget—and impose a balanced budget, so that a single effective tax rate τ finances observed outlays. The calibrated tax rate then satisfies $\tau \times (W\theta N/\text{GDP}) = \tau \times 0.6 = 0.056$, yielding $\tau = 9.3\%$, well below the statutory rate of 21%–23% but above observed contribution revenues alone.[15] The replacement ratio P is calibrated directly to match SIPA contributory pension outlays of 5.6% of GDP under both approaches: Under the balanced-budget calibration, this is achieved jointly with $\tau = 9.3\%$, so that revenues and outlays are equated by construction; under the deficit calibration, τ is set to match observed contribution revenues of 4.3% of GDP and P is set independently to match observed outlays, so the deficit of approximately 1.3% of GDP emerges as an equilibrium outcome. We set the retirement age at 65, the statutory age for men under Law 24,241 ($R = 47$ in model age; Section 2.1).[16]

4.2 Intergenerational Redistribution Under a Stationary, Balanced-Budget Social Security

This section examines who gains and who loses from SS in a hypothetical world with a balanced SS budget and a stationary age distribution. This environment, proposed by Samuelson (1975), is ideal for isolating the intergenerational transfers embedded in a well-functioning PAYG system. We extend Samuelson's original analysis by incorporating a richer demographic structure and applying it to the 2023 parameter values described in Section 4.1. As in Samuelson, the balanced-budget assumption makes the generosity ratio—the ratio of the benefit replacement rate to the payroll tax rate—an equilibrium object determined by demographics.

We focus on the SS rate of return for a newborn cohort, denoted $\hat{r}_{0,t}$, and on the break-even generation's age $\bar{a}$, for which $\hat{r}_{\bar{a},t} = 1$. Demography's role in $\hat{r}_{0,t}$ is captured by the term $\tilde{n}$ in (24'), $\hat{r}_{0,t} = ((1+\gamma)/(1+r))^{R+1}(1+\tilde{n})^{R+1}$. The demographic growth factor satisfies $(1+\tilde{n})^{R+1} = \frac{P}{\tau}\frac{\mathscr{S}}{\mathscr{L}_0}$, the product of the equilibrium generosity ratio in (14) and the ratio of the expected discounted value of a unit of retirement income to that of a unit of contributions (the ratio is a proxy for expected years in retirement relative to expected years of contributions). We report two counterfactuals:

that, when annualized, yields pension outlays of roughly 5%–6% of GDP, consistent with the official figure.

[15] The statutory contribution rate to SIPA is approximately 21%–23% of gross wages (11% employee plus 10%–12% employer). The calibrated rate of 9.3% lies between observed contribution revenues (4.3% of GDP, implying $\tau = 7.2\%$) and the statutory rate, reflecting the partial coverage of general revenue transfers. The statutory replacement rate for a worker with 30 years of contributions is 45% of average pensionable wages (1.5% per year of service under Law 24,241). The model-implied rate is substantially higher—around 58%—driven by the pension floor (*haber mínimo*), discretionary top-ups (*bonos*), and the fact that around 64% of beneficiaries accessed pensions through a *moratorium* with little actuarial link to contributions as of 2023 (ANSES, 2023), rising to 68.6% of retirement pensions by September 2025 (Ministerio de Trabajo, Empleo y Seguridad Social, 2025) (Table 7).

[16] The statutory retirement age is 65 for men and 60 for women. We use the male retirement age as the baseline. A lower R raises the old-age dependency ratio, reducing the balanced-budget generosity ratio P/τ, the return $\hat{r}_a$, and the break-even age $\bar{a}$.

the fixed-age-distribution scenario, which is backward-looking and preserves the age structure inherited from past fertility, and the fixed-fertility scenario, which is forward-looking and considers the stationary age distribution derived from 2023 fertility rates and survival probabilities, computed from equation (1) using the 2023 age distribution as the initial condition.

Table 3 reports the results. At our benchmark interest rate of 6%, SS transfers resources from the young to the old. For the fixed-fertility scenario, which projects current fertility rates forward, the youngest cohort expects to lose 82% of its contributions. For the fixed age distribution, an entrant worker loses 57%. SS favors the young—and is a Pareto improvement over laissez-faire—in only one configuration: $r = 2\%$ under the fixed age distribution, in which $\hat{r}_{0,t} = 1.68$, reflecting the favorable ratio of $(1+\gamma)$ to $(1+r)$ compounded over the $R+1$ years to retirement.

The demographic transition is consequential. Projecting current fertility forward implies a long-run population growth rate of -1% per year and cuts the contributor-to-beneficiary ratio N_t/D_t from 5.42 to 2.15, which more than halves the fiscally feasible generosity ratio P/τ, from 6.29 to 2.66. Because $\tilde{n}$ falls with it, returns are below one even at $r = 2\%$ under fixed fertility ($\hat{r}_{0,t} = 0.71$): Having fewer workers per retiree necessarily reduces what a balanced-budget PAYG system can pay.

Table 3 shows how the demographic transition translates into political support for SS's intergenerational redistribution. Although the system is unprofitable at birth in most configurations, we know from Lemma 2 that $\hat{r}_a$ increases with age, a. We compute the return conditional on survival to each age and identify the youngest worker for whom participation is welfare improving. That cutoff rises substantially over time: At $r = 6\%$ the youngest worker in favor of the system is 37 years old with the current age distribution and 50 years old in the steady state.

Despite the rising cutoff, population aging prevents opposition from becoming politically dominant. In all cases the share of individuals who benefit from the system exceeds one-half, even though the system generates substantial losses for the young. A reform eliminating the system would therefore fail under majority voting.

Table 3: Break-Even Generation and Mass in Favor

	$\bar{a}_t$			Mass $a_t \geq \bar{a}_t$			$\hat{r}_{0,t}$		
r	2%	4%	6%	2%	4%	6%	2%	4%	6%
Fixed Age Distribution (2023)	18	24	**37**	1.00	0.87	**0.61**	1.68	0.86	**0.43**
Fixed Fertility (2023)	34	45	**50**	0.80	0.65	**0.57**	0.71	0.36	**0.18**

Note: The variable $\bar{a}_t$ denotes the break-even generation defined in equation (22). Mass $a_t \geq \bar{a}_t$, is the fraction of the population at or above $\bar{a}$. SS's return at birth $\hat{r}_{0,t}$ is defined in (24').

4.3 Fiscal Sustainability Under Alternative Policies

We organize the policy analysis in three stages: the fiscal trajectory under current policy, or the business-as-usual (BAU) scenario; the adjustments to the contribution rate τ or the replacement ratio P required for fiscal sustainability, as well as their cost; and the retirement age as an additional instrument, alone and in combination.

Since public funds are fungible, no stock of existing sovereign debt can be attributed to social security alone. We therefore define the *cumulative pension debt* as the debt stock a government would accumulate by financing pension deficits entirely through borrowing, starting from a zero position in 2023—the most conservative baseline, crediting the system with no inherited liabilities. All simulations begin in 2023.

4.3.1 The Business-as-Usual Scenario

The BAU policy is calibrated to 2023 data, reproducing the observed pension deficit of 1.3% of GDP. Revenues are flat by construction: Under Cobb-Douglas technology and a fixed contribution rate, they equal $\tau(1-\alpha)$, a constant fraction of output. Outlays rise monotonically with the old-age dependency ratio, so the deficit widens from the expenditure side alone. The left panel of Figure 4 shows the deficit-to-GDP ratio under BAU and under a counterfactual that zeroes the 2023 deficit by recalibrating τ; both widen at nearly identical rates. The right panel shows the implied cumulative pension debt against the natural borrowing limit of Section 2.4.1, set at 15.9 times GDP (equation (17) with $n=0$; the model's projected long-run population decline of roughly -1% per year tightens it to about 12.7). Under BAU, debt crosses the limit by 2090; the zero-deficit counterfactual delays the crossing by about a decade. Either way, the structural imbalance from aging, not the inherited fiscal position, exhausts the system's borrowing capacity.

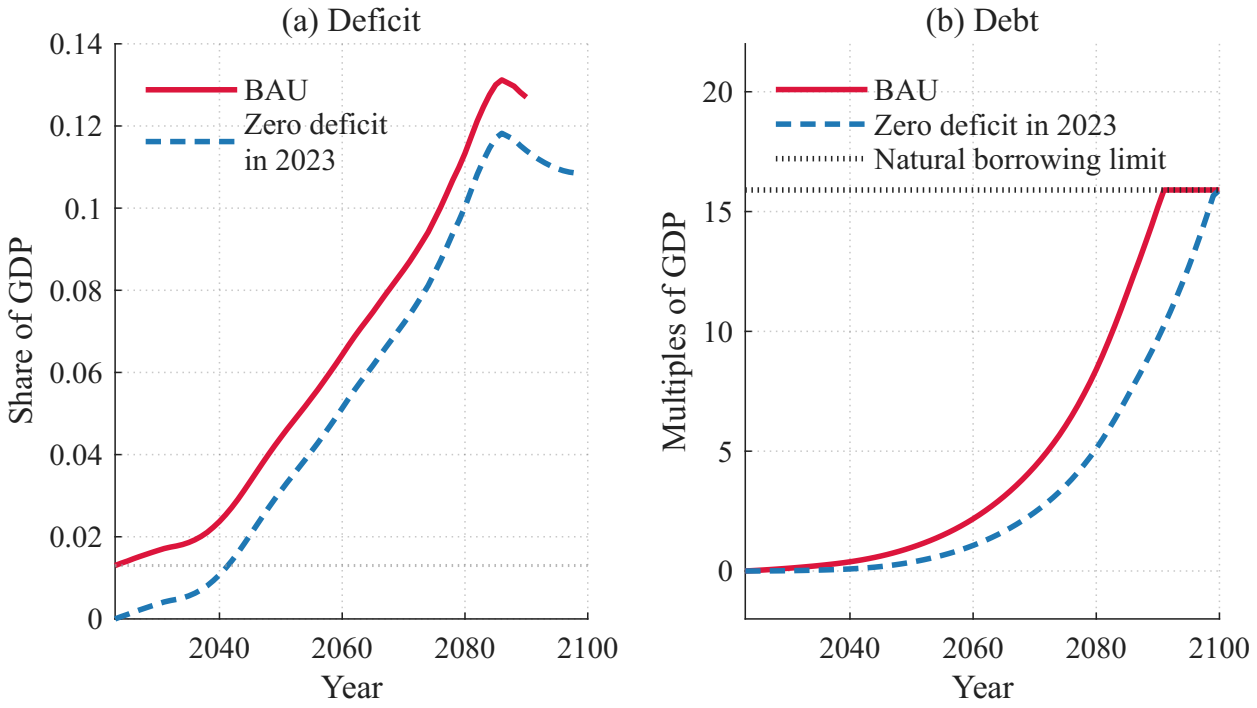
4.3.2 Fiscal Adjustment Through Contributions and Benefits

We now ask what paths of τ and P are required to achieve two alternative fiscal targets: a balanced budget ($d_t = 0$ for all t) and a deficit-to-GDP target of 2% of output.[17] For each target, the required instrument path is obtained directly from the flow budget balance: setting $d_t = \delta_t Y_t$ and solving for the adjusting instrument, as described in Section 2.4.

Figure 5 reports the results. To facilitate economic interpretation, the left panel expresses the required adjustment in terms of pension revenues as a share of GDP, equal to $\tau_t(1-\alpha)$, rather

[17]The target is chosen simply as a reference point rather than a policy recommendation; it sits above the 1.3% deficit actually observed in 2023. We could also consider a debt-to-GDP ceiling as a fiscal rule, but it implies a discrete upward jump in τ or a discrete downward jump in P at the date the ceiling binds, the magnitude of which increases with the level of the ceiling. We therefore focus on the two flow-based rules, which imply smooth adjustment paths and serve as cleaner benchmarks.

Figure 4: Deficit and Cumulative Pension Debt



Note: Left panel shows deficit-to-GDP ratio under business-as-usual (BAU) and under a zero-deficit counterfactual. Right panel shows the implied cumulative pension debt starting from zero in 2023, alongside the natural borrowing limit from equation (17).

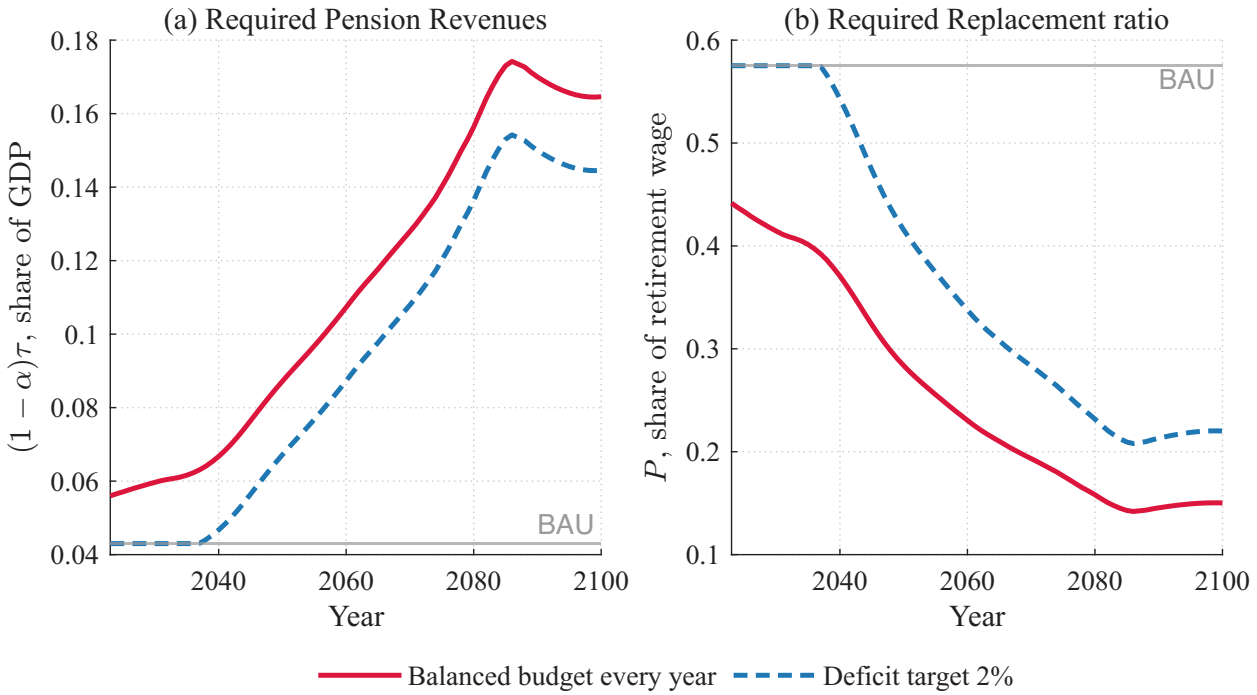
than the model contribution rate τ_t directly.[18] The right panel shows the required replacement ratio P_t under the same rules.[19] Achieving a balanced budget requires pension revenues to rise from 4.3% of GDP in 2023 to 16.5% of GDP by the end of the century. Whether this is achieved through a higher statutory contribution rate, a broader contribution base, reduced informality, or reallocation from other fiscal revenues is a policy-design question beyond the scope of the model. What the model establishes is the required scale of revenue mobilization.

An equivalent adjustment through the replacement ratio is equally dramatic: The average

[18]The model's contribution rate τ is calibrated to match observed pension revenues as a fraction of the aggregate wage bill and does not correspond to the statutory payroll tax rate, which stands at approximately 21%–23% in the data. Mapping τ_t onto a statutory rate requires assumptions about informality, compliance, and the composition of the tax base that are beyond the scope of this model. The revenue-to-GDP representation is instrument neutral: It states the required scale of revenue mobilization without prescribing how it is achieved.

[19]The model's replacement ratio P is calibrated to match aggregate pension outlays as a fraction of the wage bill and is best interpreted as an average effective replacement ratio across all beneficiaries, weighted by cohort size. It does not correspond directly to the statutory replacement rate, which varies by contribution history, benefit regime, and the presence of moratoria. The interpretation of changes in P is therefore analogous to that of τ: The model prescribes the required change in average effective benefit generosity; the implementation through specific statutory instruments is a separate policy-design question.

Figure 5: Required Fiscal Adjustment Under Alternative Rules



Note: Left panel shows required pension revenues as a share of GDP, equal to $\tau_t(1-\alpha)$, under a balanced-budget rule (solid) and a deficit-to-GDP target (dashed). The horizontal dotted line marks the business-as-usual (BAU) revenue level. Right panel shows the required average effective replacement ratio P_t under the same rules. The debt path is identical across instruments for a given fiscal rule; see main text.

effective replacement ratio must fall from 58% to about 15% by 2100. A replacement ratio of 15% would provide only minimal income support, effectively amounting to a de facto phase-out of pension generosity.

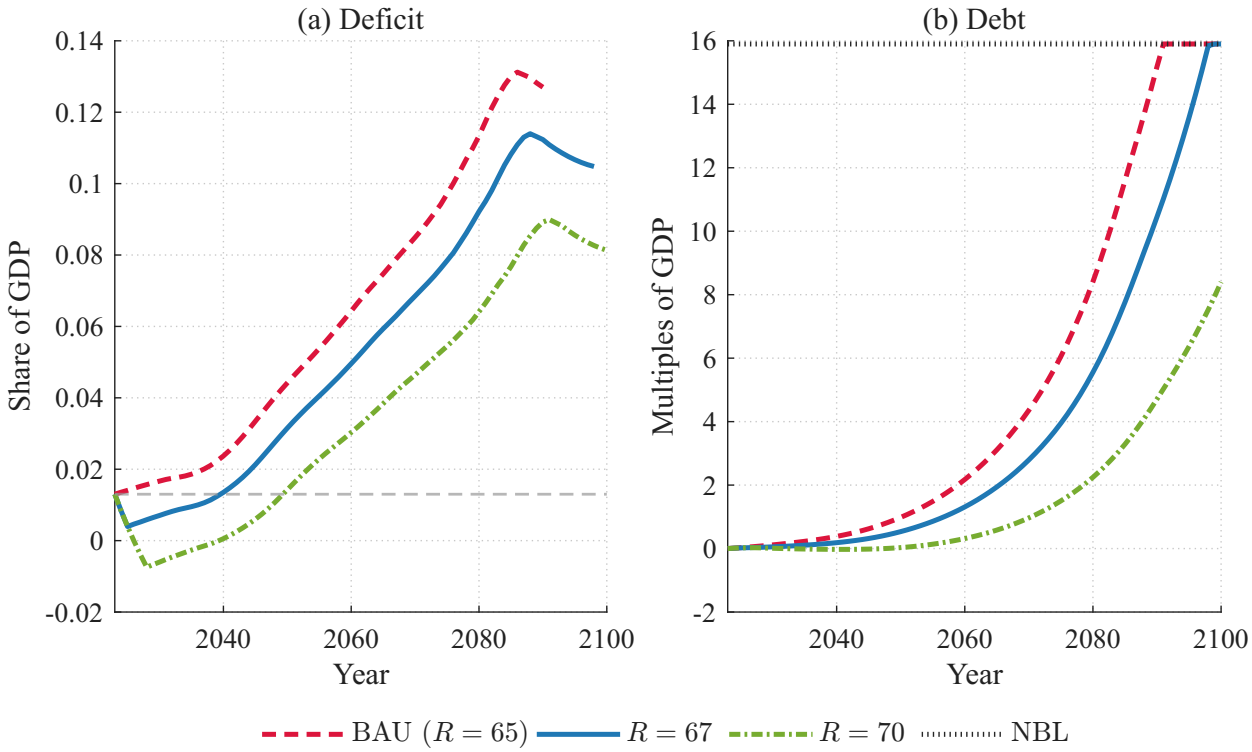
The debt path implied by a given fiscal rule is independent of which instrument—τ or P—is used to achieve it: The flow budget constraint pins down d_t, which governs debt accumulation via equation (15a), irrespective of how the required adjustment is distributed between workers and retirees. The instrument choice is therefore a distributional decision, not a fiscal one. Under the deficit-target rule, maintaining a 2% deficit-to-GDP target slows debt accumulation relative to BAU but does not prevent it.

4.3.3 Adjusting the Retirement Age

Figure 6 examines whether extending the retirement age alone can restore fiscal sustainability. Holding τ and P at their BAU values, we simulate the system under $R = 67$ and $R = 70$. Both scenarios reduce the deficit: A higher retirement age shrinks the dependent population D_t, reducing

outlays, while simultaneously expanding the working population N_t, increasing revenues. The deficit path shifts down and the debt trajectory flattens relative to BAU. Neither extension prevents an eventual violation of the natural borrowing limit, however.

Figure 6: Fiscal Dynamics Under Retirement-Age Reform



Note: Left panel shows deficit-to-GDP under business-as-usual (BAU) ($R = 65$, dotted crimson), $R = 67$ (solid blue), and $R = 70$ (solid green). Right panel shows the corresponding cumulative pension debt paths alongside the natural borrowing limit (NBL) (dashed dark). All series start from zero in 2023.

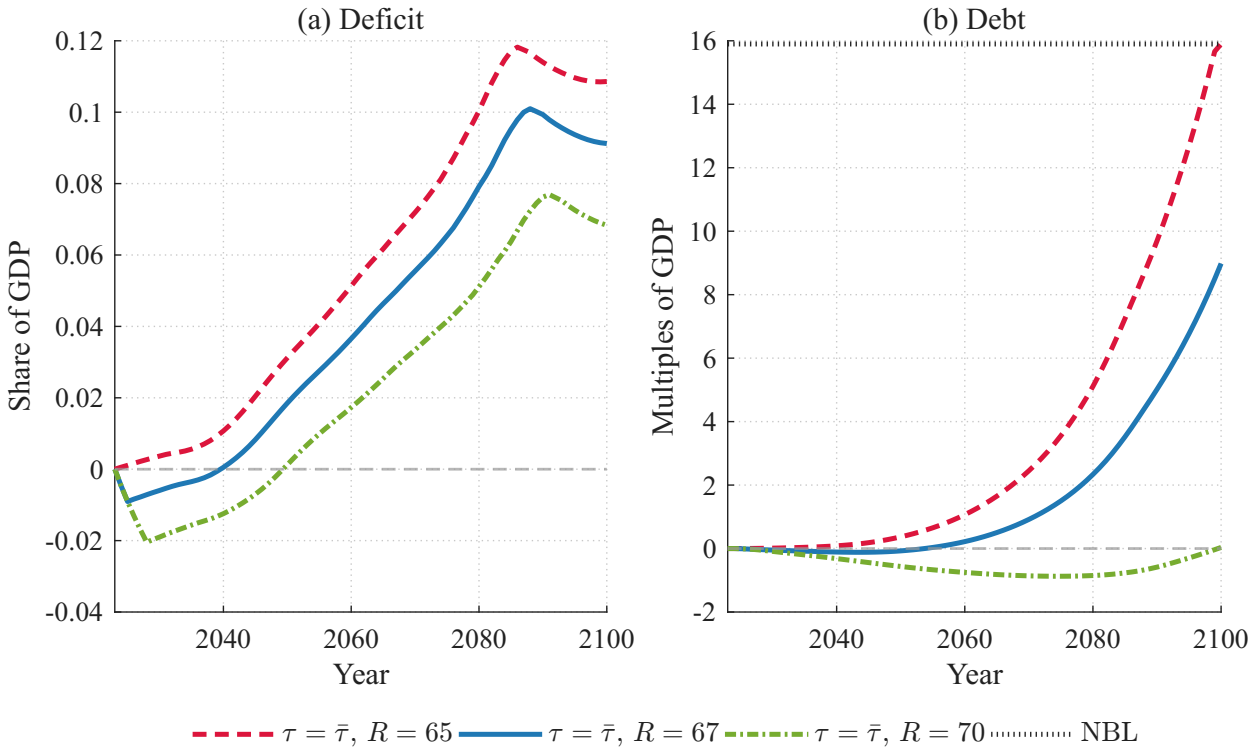
Two limitations bound the retirement-age analysis. First, computing the fiscal and welfare effects of a higher R requires efficiency-unit profiles θ_h beyond age 65, at which point direct estimates of age-earnings profiles become unreliable. We construct θ_h from the life-cycle wage-growth estimates for developing countries in Lagakos et al. (2018).[20] Second, the model abstracts from any disutility of labor: With inelastic labor supply, extending the retirement age is mechanically welfare improving for any cohort whose labor income exceeds pension income. Section 4.4 addresses this directly: We compute the welfare and distributional effects of reforms that change the retirement age, augmenting the model with an explicit disutility of work.

[20]We use their Heckman-Lochner-Taber decomposition, which separates the pure return to experience from cohort and aggregate wage-growth effects and, under its own identifying restriction, is flat over the last decade of the working life—so no extrapolation beyond the estimation sample is required to extend the profile through age 70.

4.3.4 A Combined Reform

The preceding results establish two partial findings: closing the current deficit through τ or P alone requires large adjustments, and extending the retirement age alone is insufficient. We now combine both instruments, defining $\bar{\tau}$ as the contribution rate that eliminates the 2023 deficit and holding it fixed thereafter, and consider retirement ages $R = 67$ and $R = 70$.

Figure 7: Fiscal Dynamics Under the Combined Reform



Note: Left panel shows deficit-to-GDP under $\bar{\tau}$ and $R = 65$ (dotted crimson), $\bar{\tau}$ and $R = 67$ (solid blue), and $\bar{\tau}$ and $R = 70$ (solid green), where $\bar{\tau}$ is the contribution rate that closes the 2023 deficit. The right panel shows the corresponding cumulative pension debt alongside the natural borrowing limit (NBL) (dashed dark). All series start from zero in 2023.

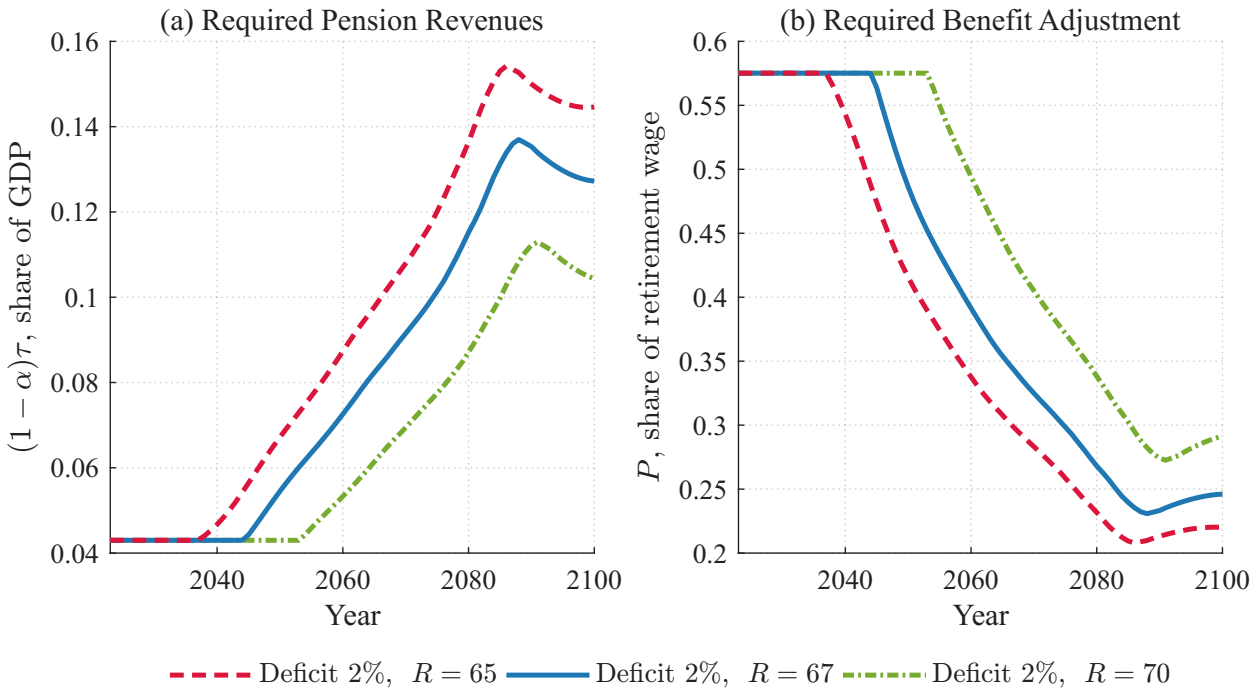
Figure 7 shows the resulting fiscal dynamics. The combination generates primary surpluses for approximately 16 years under $R = 67$ and 26 years under $R = 70$: During this period the pension system accumulates assets rather than debt. Sustained demographic pressure eventually reverses the surplus in both cases, though at very different paces: Under $R = 67$ the system re-enters net debt by 2055, while under $R = 70$ it remains asset positive until nearly the end of the century.[21]

[21] The rise in debt lags the reappearance of the deficit: Since $r > \gamma + n$ throughout, the accumulated asset buffer continues to compound in magnitude on its own, and this effect must be outweighed by the recurring deficit before debt-to-GDP visibly turns the corner—a lag that is longer the larger the buffer (i.e., longer under $R = 70$ than under $R = 67$).

The key distinction from single-instrument reforms is the extended fiscal runway: The system has substantially more time before solvency constraints bind.

The combined reform depicted in Figure 7 requires an immediate increase in contributions to close the 2023 deficit. A less front-loaded alternative asks how large an ongoing adjustment would be needed if, instead, the government simply committed to a 2% deficit-to-GDP ceiling going forward, without first balancing the budget. Figure 8 answers this. Under this 2% target, extending the retirement age reduces the required increase in pension revenues relative to no reform—by 1.7 percentage points under $R = 67$ and by 4.0 percentage points under $R = 70$. The required benefit reduction is proportionally smaller still. Extending the retirement age also buys time before the ceiling binds: 18 years under $R = 67$ and 28 years under $R = 70$.

Figure 8: Required Ongoing Adjustment Under the Combined Reform

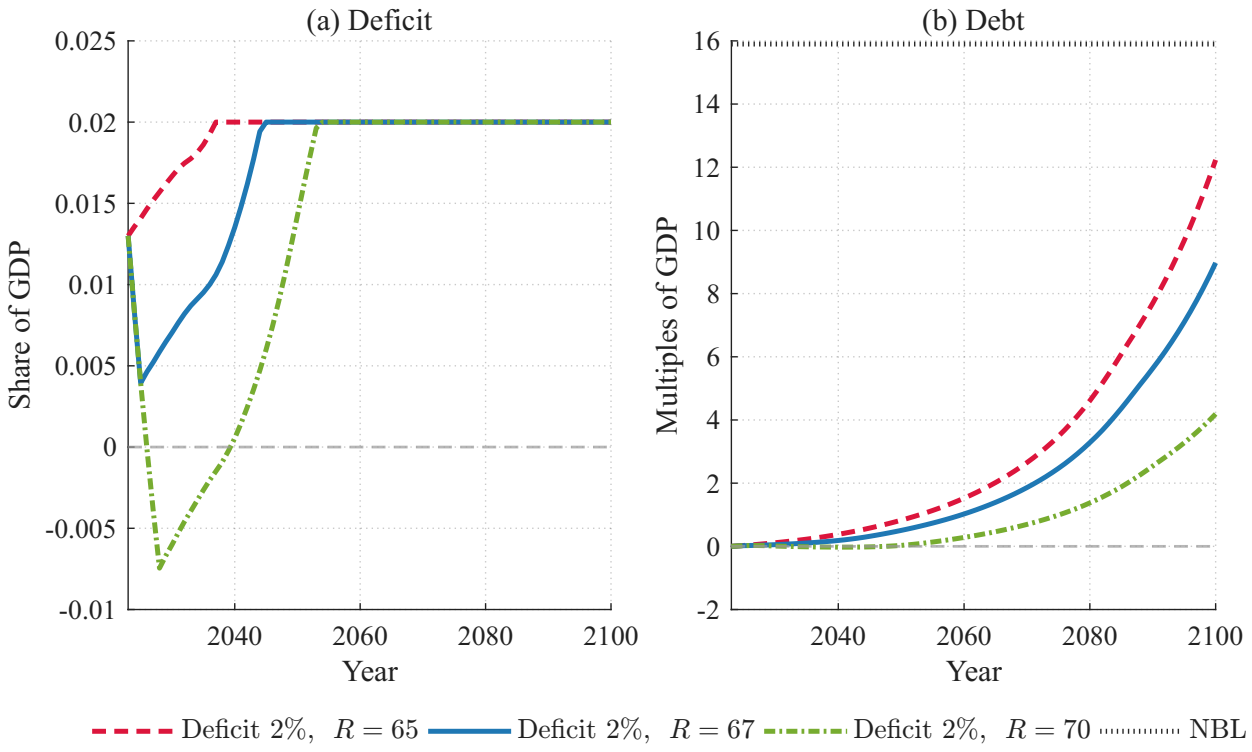


Note: Left panel shows required pension revenues as the share of GDP, equal to $\tau_t(1-\alpha)$, needed to keep the deficit below 2% of GDP under $R = 67$ (solid blue) and $R = 70$ (dash-dot green). The dashed crimson line shows the same required adjustment under $R = 65$ (no retirement-age reform), for reference. Right panel shows the required replacement ratio P_t under the same scenarios.

Figure 9 shows that without the upfront revenue adjustment, neither retirement age generates the sustained surplus of Figure 7. The retirement-age reform pushes the deficit below the 2% ceiling for a period—down to about 1% under $R = 67$ and to nearly zero under $R = 70$—but not into an actual surplus, so cumulative debt keeps rising throughout the horizon under both scenarios

but more slowly when the ceiling is slack. The larger reform simply buys a longer and deeper window before the ceiling binds again: It stays slack until 2040 under $R = 67$ and until 2050 under $R = 70$.

Figure 9: Deficit and Debt Under the Combined Reform with Deficit Commitment



Note: Left panel shows deficit-to-GDP under $\bar{\tau}+R=67$ (solid blue) and $\bar{\tau}+R=70$ (solid green) when the government commits to a 2% deficit-to-GDP ceiling. Right panel shows the corresponding cumulative pension debt alongside the natural borrowing limit (NBL) (dashed dark). Negative debt denotes asset accumulation by the pension system.

Taken together, the results point to two complementary levers rather than a single prescription. Closing the 2023 deficit through a modest contribution increase and raising the retirement age to $R = 70$ yields nearly three decades of asset accumulation (Figure 7); alternatively, raising the retirement age under a 2% deficit ceiling, with no upfront adjustment, cuts the eventual required revenue increase by more than a third (Figures 8 and 9). Either way, the retirement age is the lever that buys the most runway. Whether such a path is welfare improving and politically supportable depends on how its costs fall across cohorts—the question asked in the next section.

4.4 The Distributional Consequences of Reform

Section 4.3 shows that the PAYG system will eventually hit the natural borrowing limit; Section 4.2 shows that it already pays young workers a negative return. The fiscal adjustments compound that loss: Raising τ taxes working-age cohorts without improving their benefits, and cutting P reduces benefits for cohorts that contributed at the old rates. Either way, preserving the system comes at the expense of those who were already its net losers.

Table 4 documents this across four scenarios: BAU; a one-off adjustment that zeroes the 2023 deficit and is then held fixed;[22] and two ongoing rules that hold the deficit at 2% of GDP thereafter, one through τ and one through P. Under business-as-usual, the implicit return at birth is already deeply negative and only cohorts above a certain age support the system. Every fiscal adjustment pushes the break-even age higher and shrinks the share of the population that gains from the system being preserved—the opposite of what a politically sustainable reform should do.

Table 4: Break-Even Generation, Mass in Favor, and Return to Social Security under Alternative Fiscal Scenarios

	$\bar{a}_t$ (youngest in favor)	**Mass** $a_t \geq \bar{a}_t$ (% of pop.)	$\hat{r}_{0,t}$
BAU	29	77.1	0.68
Zero deficit in 2023	35	64.9	0.52
Deficit target 2% with τ	32	70.9	0.52
Deficit target 2% with P	39	57.1	0.29

Note: $\bar{a}_t$ is the age of the youngest cohort with a non-negative return on SS participation. Mass $a_t \geq \bar{a}_t$ is the share of the living population at or above that age. $\hat{r}_{0,t}$ is the implicit rate of return accruing to a newborn under each policy scenario: business-as-usual (BAU); a one-off adjustment that zeroes the 2023 deficit and is then held fixed; and two ongoing rules, adjusting τ or P, that target a 2% deficit-to-GDP ceiling thereafter. The zero-deficit-in-2023 row gives the same statistics whether τ or P does the adjusting since both reach the same generosity ratio P/τ.

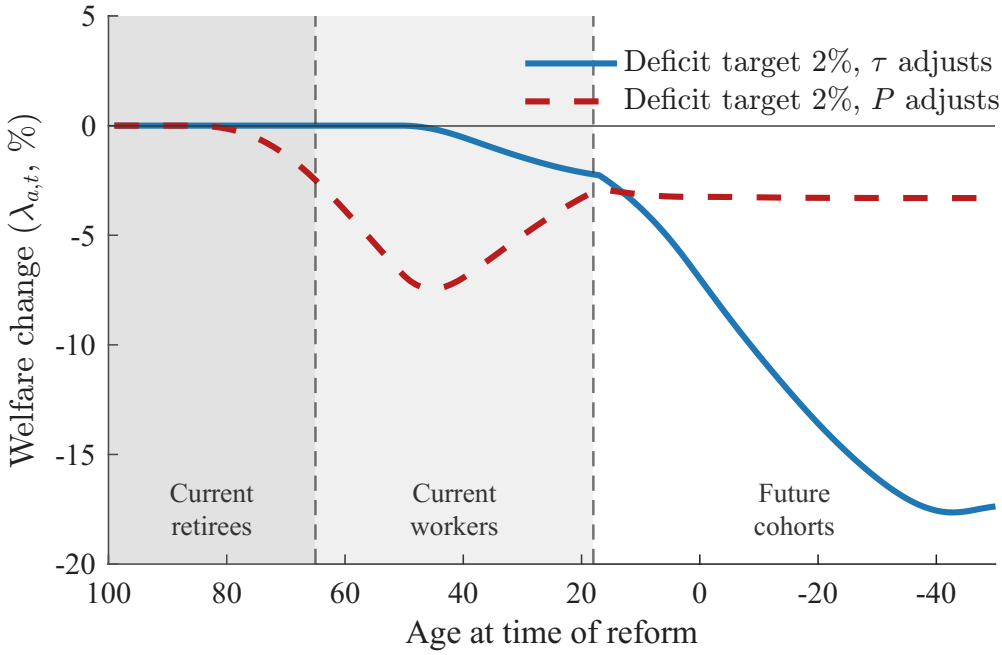
Table 4's BAU return is substantially higher than the stationary benchmarks of Section 4.2: P there is calibrated to reproduce the observed 2023 deficit rather than to balance the budget, so it embeds the extra generosity that debt financing allows. The fiscal adjustments that follow remove that slack. Both instruments considered below also share an assumption worth stating plainly: Pension promises already made are not honored. A benefit cut under P adjustment falls on people who are already retired, exactly as it falls on new retirees: Their pension is not protected at the

[22] This row does not reproduce Section 4.2's stationary benchmarks. It balances the budget only in 2023, using that year's still-favorable demographics, and then holds τ and P fixed as the population continues to age, so deficits reopen and debt accumulates over the newborn's lifetime, just as under BAU, but from a lower starting point. Section 4.2, by contrast, evaluates a frozen hypothetical—either 2023 demographics or the converged steady state—held forever. Both differences push this row's implied return above both stationary benchmarks at $r = 6\%$.

level they retired on.

Figure 10 shows the full cohort profile of consumption-equivalent welfare changes $\lambda_{a,t}$ relative to BAU. Current retirees are unaffected under τ adjustment but lose under P adjustment. Under τ adjustment, losses deepen monotonically into the future, with the largest losses falling on cohorts not yet born. Under P adjustment, losses are hump shaped, peaking not among the young but among workers in early middle age.[23] The figure also shows an additional result: τ adjustment is strictly worse for younger cohorts and future generations than P adjustment. This is not coincidental. When the system offers a negative return, reducing benefits moves it closer to a phase-out—the outcome young workers would prefer—while raising contributions deepens their forced participation in a loss-making arrangement. Conditional on preserving the system, P adjustment is the least harmful option for the young precisely because it shrinks the transfer they are compelled to make.

Figure 10: Welfare Change by Cohort Under Alternative Fiscal Adjustments



Note: Consumption-equivalent welfare change $\lambda_{a,t}$ relative to business-as-usual. The τ adjustment (blue solid) raises contributions to target a 2% deficit-to-GDP ceiling; the P adjustment (crimson dashed) reduces the replacement ratio to the same target. Vertical lines delimit current retirees, current workers, and future cohorts.

[23] The hump reflects household balance sheets, not pension arithmetic. With complete markets and a rising wage profile, agents borrow heavily in early adulthood against future labor and pension income; the status quo asset position is negative for most of the working life and is only gradually reduced as retirement approaches. A given cut in expected pension income is therefore a larger proportional hit to net wealth for the cohorts still carrying substantial debt than for either the very young, who have not yet borrowed against future income, or workers close to retirement, whose debt has been mostly repaid.

4.4.1 *Raising the Retirement Age: Accounting for Disutility of Work*

The comparisons above measure welfare by consumption alone. That is adequate for changes in τ and P, which leave working life unchanged, but not for a higher retirement age: Extra working years cost leisure, home production, and effort, none of which the model's inelastic labor supply counts, so the model would mechanically favor the reform for any cohort whose labor income exceeds pension income. To make the comparison consistent across reforms, we augment the analysis with a disutility of work.

Setup. Suppose the per-period utility of an agent of age a is $u(c_a) - \psi \cdot \mathbf{1}[a \leq R]$, where $\psi \geq 0$ is the per-period disutility of working. We read ψ broadly, as encompassing both the value of the leisure and home production time that working displaces and the work-related expenditures that the model counts as consumption but that yield no utility; both enter the welfare calculation in the same way. Rather than assigning an arbitrary utility value to ψ, we express it in units with an empirical counterpart: a fraction x of the agent's certainty-equivalent consumption $\bar{c}_a$,

$$\psi \;=\; x \cdot u'(\bar{c}_a)\,\bar{c}_a. \tag{32}$$

Consider then extending the retirement age by $k > 0$ years, so that $R' = R + k$. Under CRRA preferences, this yields a net consumption-equivalent welfare gain

$$\left(1 + \lambda^{\text{net}}_{a,k}(x)\right)^{1-\sigma} \;=\; \left(1 + \lambda_{a,k}\right)^{1-\sigma} \;-\; x(1-\sigma) \cdot \omega_{a,k}, \tag{33}$$

where $\lambda_{a,k}$ is the consumption-equivalent welfare gain defined in equation (18), $\widetilde{T}_{a,t} \equiv \sum_{h=0}^{T-a} \beta^h S_{a+h,t+h}/S_{a,t}$ is the discounted remaining lifetime, $\widetilde{T}^{w}_{a,k} \equiv \sum_{j=\max(R+1-a,0)}^{R+k-a} \beta^j S_{a+j,t+j}/S_{a,t}$ is the discounted number of extra working periods the reform imposes, and $\omega_{a,k} \equiv \widetilde{T}^{w}_{a,k}/\widetilde{T}_{a,t}$ is the share of an agent's discounted remaining life the reform converts into work.[24] When $\lambda_{a,k}$ is small, (33) is well approximated by the more transparent linear form $\lambda^{\text{net}}_{a,k}(x) \approx \lambda_{a,k} - x \cdot \omega_{a,k}$; we use the exact expression above throughout since $\lambda_{a,k}$ is not always small in our calibration.[25]

Setting $\lambda^{\text{net}}_{a,k}(x) = 0$ defines *break-even disutility*

$$x^*(a,k) \;=\; \frac{\left(1 + \lambda_{a,k}\right)^{1-\sigma} - 1}{(1-\sigma)\,\omega_{a,k}}, \tag{34}$$

[24] For agents already retired when the reform is announced—those aged $a \geq R$ at $t = 0$—we set $\widetilde{T}^{w}_{a,k} = 0$ and $\omega_{a,k} = 0$. For subsequent periods, $\widetilde{T}^{w}_{a,k}$ is nonzero only for cohorts who had not yet retired at the reform date.

[25] In the linear approximation, $x \cdot \omega_{a,k}$ is the utility cost of the reform, expressed in consumption-equivalent units: It is zero for current retirees, who face no extra work, and grows as a approaches the original retirement age R, where the extra working years the reform imposes make up a larger share of remaining life.

the value of x above which agent a prefers the status quo. The disutility $x^*(a,k)$ decreases with age, both because $1/\omega_{a,k}$ shrinks as agents approach R and because of the curvature of the utility function: It bites hardest on the largest welfare gains, which accrue to cohorts closest to retirement. The reform is most fragile among cohorts near the original retirement age R. See Appendix F for the full derivation.

Table 5: Break-Even Disutility of Work, by Retirement-age Reform

R'	$x^*(65,k)$
66	0.452
67	**0.416**
68	0.387
69	0.363
70	0.343

Note: $x^*(65,k)$ is the disutility that leaves a 65-year-old indifferent between the status quo and a one-time jump to $R' = 65+k$ (equation (34)). The calibration target used throughout is $x^*(65,2) = 0.416$ (bold row).

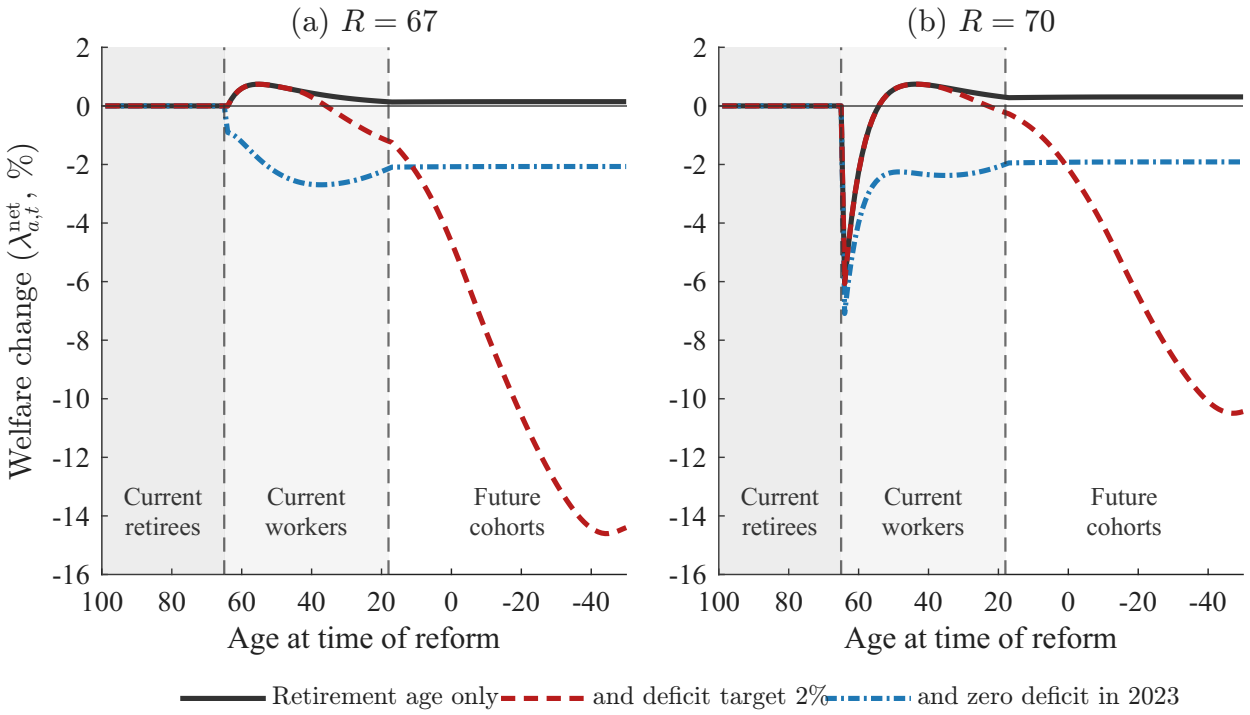
Parametrization of x. We calibrate x using the model's own retirement margin. The reforms considered below raise the retirement age in one step, from 65 to R', so we ask what disutility of work would leave a 65-year-old exactly indifferent between the status quo and that jump. Equation (34), evaluated at $a = 65$, gives this one-shot break-even threshold $x^*(65,k)$ for $R' = 65+k$.

Table 5 reports $x^*(65,k)$ for each reform considered, $R' = 66$ through 70. We set x to the value that would leave a 65-year-old exactly indifferent to the $R' = 67$ reform: $x^*(65,2) = 0.416$.

Results. Figure 11 shows the welfare effect of the retirement-age reform, alone and paired with a contribution adjustment, at the calibrated $x = 0.416$, for $R' = 67$ (left) and $R' = 70$ (right). Current retirees are unaffected in every variant: Nothing changes their retirement age or pension. The reform alone is close to costless: No worker opposes $R' = 67$, and only workers within a decade of today's retirement age oppose $R' = 70$ (13.4% of workers). Pairing the reform with a permanent 2% deficit target preserves majority support among workers—52.7% in favor under $R' = 67$ and 69.1% under $R' = 70$, with the higher retirement age requiring the smaller ongoing adjustment. Financing it instead with a one-off tax increase that balances the 2023 budget reverses this almost entirely: Essentially every current worker and future cohort opposes both reforms.

Read together, these results point to a single conclusion: The reforms that come closest to restoring fiscal sustainability are the furthest from welfare improving, particularly for cohorts not yet born. The retirement-age extension alone is nearly costless; paired with a permanent deficit ceiling, it does part of the fiscal work, so the required ongoing τ or P adjustment is smaller than

Figure 11: Welfare Change Under Retirement Age Reform, With and Without a Contribution Adjustment



Note: Consumption-equivalent welfare change $\lambda^{\text{net}}_{a,t}(x)$ relative to business-as-usual, at the calibrated $x = 0.416$. Panels are $R' = 67$ (left) and $R' = 70$ (right). Each panel compares the retirement-age reform alone to two ways of financing it with a contribution adjustment: a one-off increase that balances the 2023 budget, and a permanent commitment to a 2% deficit-to-GDP target. Vertical lines delimit current retirees, current workers, and future cohorts.

it would be without it—but almost all of its cost falls on cohorts who cannot yet vote. The next section turns to the electorate that would have to approve any of these reforms.

5. The Current Politics of Social Security

In the previous sections, we studied the intergenerational distributive consequences of social security and reported the mass in favor of the PAYG system using our estimates of the population's age distribution. This section studies the politics of SS by comparing the break-even age with the actual age distribution of voters. A PAYG system that offers a deeply negative implicit return to the young survives only where participation is compulsory; its political viability therefore hinges on the share of voters above the break-even age and on how that share shifts as the population ages.

Table 6 summarizes the age distribution of the population and reported voters in the 2023

Table 6: Voting by Age Group, 2023

Age Group	Total Population	Reported Voters	Participation (%)
Total 18 years & over	34,180,015	26,870,779	78.6
Median age (Argentina)	41	41	—
Median (provincial median age)	40	40	—

Source: Data provided by the Cámara Nacional Electoral for the 2023 general election.

general presidential election. The median voter is 41 years old, the same as the median age of the adult population, consistent with compulsory voting. The last row reports the median across provinces of the province-level median voter's age, the distribution relevant for representation in the Senate.

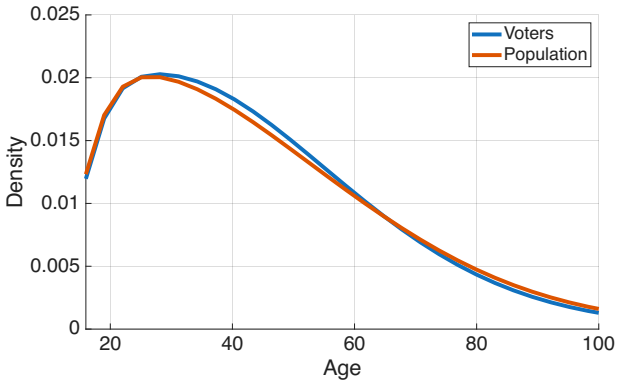
Figure 12 illustrates these patterns graphically. Panel (a) shows the 2023 age distribution of the total population and voters, confirming that the two are not very different, consistent with voting being compulsory in Argentina. Panel (b) displays the distribution of province-level median ages, which is relevant for Senate representation. Panels (c) and mainly (d) document the aging of the electorate.

Comparing the break-even age of a balanced-budget SS for the 2023 age distribution at the benchmark interest rate $r = 6\%$—37 years—with the median voter's age of 41 (Table 6) shows that a proposal to eliminate the system would lose under majority rule today. The demographic transition does not overturn this: In the steady state the break-even age rises to 50, and the mass above it—57%, down from 61% today but still a majority—continues to block reform. The intergenerational redistribution embodied in SIPA is politically self-sustaining.

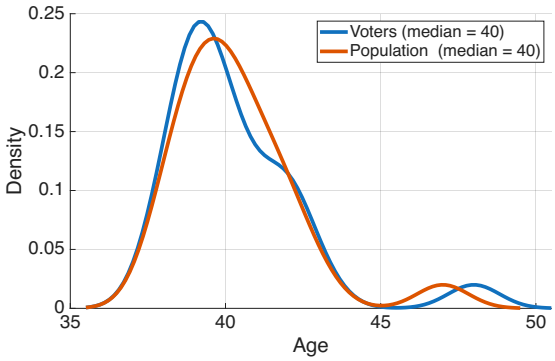
6. Final Remarks

This paper uses a tractable overlapping-generations model with uncertain lifetimes, calibrated to Argentine demographics and institutions, to address three questions about SIPA. Who gains and who loses from the current arrangement? What does it take to keep the system fiscally afloat as the population ages? And why does a system that expropriates the young nonetheless survive politically? The answers are sharp. Under any plausible benchmark for the opportunity cost of payroll contributions, SIPA delivers a deeply negative implicit return to the young. Sustaining the system requires either almost quadrupling pension revenues or cutting the average effective replacement ratio to roughly a quarter of its current level—or, more realistically, moderate revenue and benefit adjustments combined with a higher retirement age. And the electorate is sufficiently old, and ag-

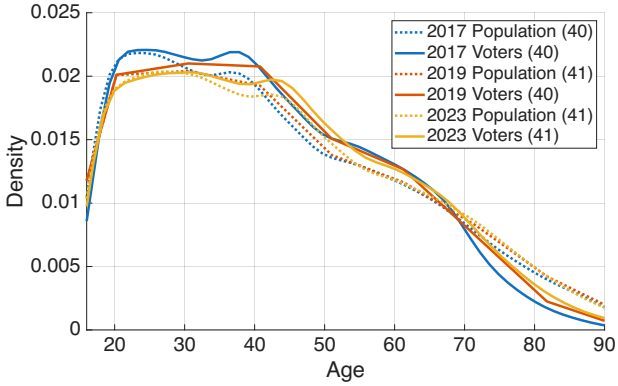
Figure 12: Age Distribution of Voters



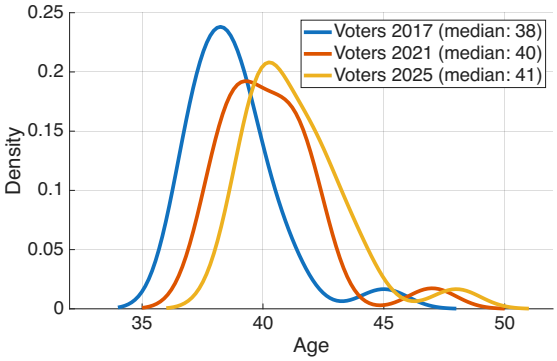
(a) 2023 Argentina Age Distribution



(b) Distribution of Province Median Age: 2023



(c) Population and Voter Age Distributions, 2017, 2019, and 2023 Elections



(d) Distribution of Province Median Voter Age, 2017, 2021, and 2025 Elections

Note: Densities are smoothed with Matlab's ksdensity function.
Source: Data provided by the Cámara Nacional Electoral.

ing sufficiently fast, that the political coalition in favor of preserving the status quo is large today and will be larger tomorrow. On the other hand, the fiscal case for reform is overwhelming on a one-generation horizon.

We close by spelling out the qualifications and extensions that the framework points toward but does not itself develop.

(i) *Distortionary taxation and the size of the pie.* With inelastic labor supply, SIPA only redistributes a fixed output across cohorts. With elastic labor supply and human-capital accumulation, the payroll tax distorts work and education decisions, and phasing out PAYG pensions raises aggregate output as well as redistributing it (McGrattan and Prescott, 2017; Daruich and Fernández, 2024; De Nardi et al., 1999). This margin would strengthen the welfare case for phase-out and shrink the losses of the cohorts that appear to lose from such a transition in our calibration.

(ii) *Fertility, altruism, and the Ponzi feedback loop.* The model treats fertility as an exogenous primitive. When old-age insurance is one motive for childbearing, a generous PAYG system depresses fertility—theory in Boldrin et al. (2015), evidence in Hohm (1975) and Rossi and Godard (2022)—and Spears and Geruso (2025) show that the welfare gains from eliminating PAYG grow once the implied fertility recovery is internalized. In a country whose fertility rate fell by 0.85 children per woman in eight years, this feedback is not hypothetical.

(iii) *Informality, formalization, and the barrier of negative returns.* About 40% of Argentine workers contribute nothing to SIPA, and 68.6% of new retirements arrive through moratoria with weak actuarial links to contributions. Formalizing informal workers could in principle close the fiscal gap (Macera and Neumeyer, 2024), but they have little incentive to enroll: SIPA's return to the young is deeply negative, the informal sector is disproportionately young (Appendix B), and non-contributory programs such as the Universal Pension for the Elderly (Pensión Universal para el Adulto Mayor) sharpen the disincentive (Galiani and Weinschelbaum, 2012; Busso et al., 2012; Levy, 2008).

(iv) *Heterogeneity within cohorts.* The representative-agent structure summarizes heterogeneous individuals by a single age. The distinctions between formal and informal workers, contributory and moratorium retirees, and the special regimes of Table 8—average benefits range from the minimum pension to more than 10 times that figure—are quantitatively of the first order. Modeling them would not change the aggregate fiscal arithmetic but would sharpen the distributive analysis and might identify reforms that are progressive within cohorts even when regressive across them.

(v) *Demographic projections.* Our projections freeze the 2023 primitives. Continued declines in fertility, further gains in longevity, or net out-migration would tighten every constraint we study; sustained immigration, which Argentina has historically attracted from neighboring countries, would slow the deterioration but probably not reverse it.

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A. The Argentine Social Security System

The social security system in Argentina is a pay-as-you-go system, where pensions and benefits are financed through payroll contributions from workers and firms. ANSES (Administración Nacional de la Seguridad Social, or National Social Security Administration) manages the Argentine Integrated Pension System (SIPA), which covers[26] the vast majority of benefits granted. The general regime to access retirement requires that the individual has reached 60 years of age (for women) or 65 (for men), and has accumulated over 30 years of contributions. If the required contributions are not met, individuals may still retire voluntarily either by completing the missing contributions, or by joining a moratorium.

Additionally, within SIPA, there is a set of special regimes designed for agricultural workers, teachers, military personnel, members of the Coast Guard, Gendarmerie or Police, construction workers, domestic workers, aviation personnel, members of the Judiciary and the Public Prosecutor's Office, scientific and technological researchers, and vineyard workers. Each of these regimes has different eligibility requirements and typically also differences in the benefits received.

As shown in Table 7, in September 2025 there were 6,106,056 beneficiaries of pensions and retirement benefits, and a total of 8,716,224 benefits granted.[27] Of the total benefits, 83.3% correspond to contributory benefits, while the remaining 16.7% correspond to non-contributory pensions, such as the Universal Pension for the Elderly (PUAM[28]) or disability pensions. Of total contributory benefits, 78% correspond to retirement pensions and 22% to other pensions. 68.6% of retirement pensions were granted through moratoria, while for pensions this figure was 32.8%. Figure 13 illustrates how the composition evolved over time. The 6.1 million SIPA beneficiaries are supported by 12.8 million formal workers in the administrative records: 10 million employees (6.2 million of them in the private sector) and 2.8 million self-employed (monotributo and autónomos). The household survey, which covers the 31 largest urban agglomerations, reports 7.9 million formal workers (Table 10 in Appendix B).

Within the total amount of benefits granted, the general regime under Law 24,241 accounts for the vast majority, 93.5% (5,309,806 benefits); special regimes account for a further 4.0% (228,164), with the remainder split between laws prior to Law 24,241 (1.8%) and provincial police and prison services (0.7%). Among special regimes, non-university teachers account for 80.0% and energy-sector workers for 7.6%, with the rest split among university teachers, scientific researchers, the Judiciary, YCF Río Turbio, and foreign service personnel. As shown in Table 8, there is substantial heterogeneity in the average benefits received across regimes.

[26] Sistema Integrado Previsional Argentino

[27] The number of benefits exceeds the number of beneficiaries because a single individual may receive more than one type of benefit.

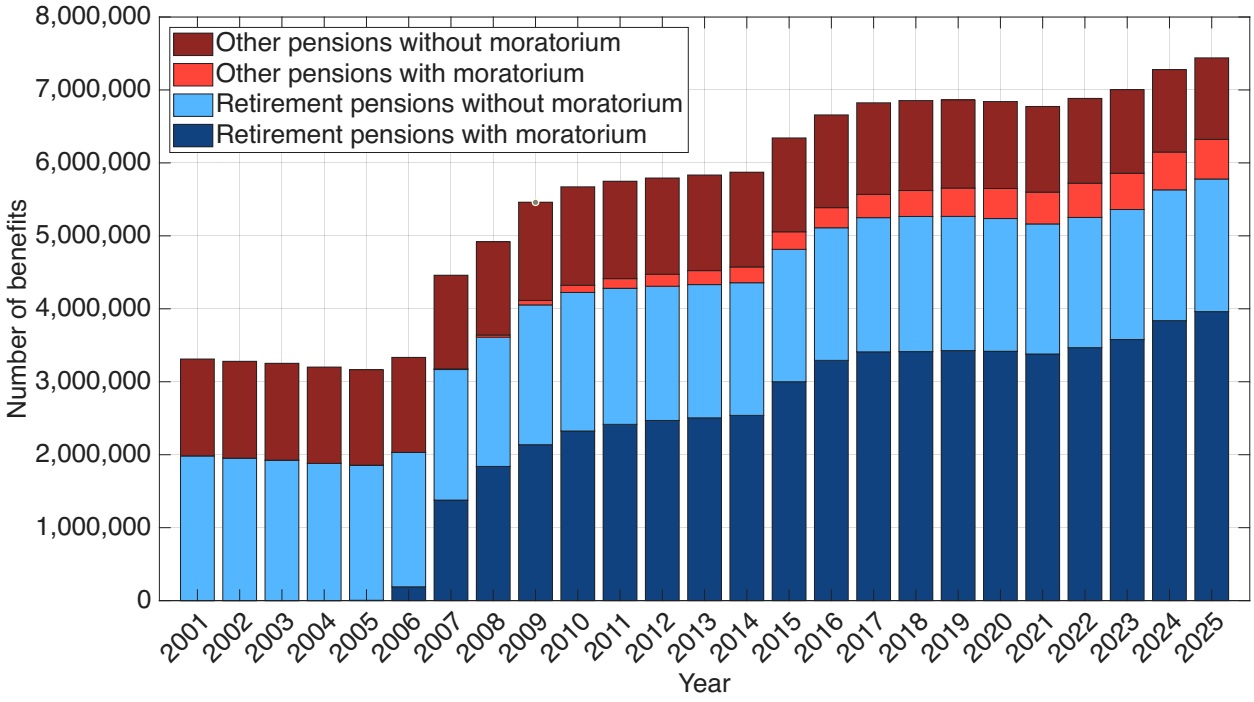
[28] Pensión Universal para el Adulto Mayor

Table 7: Number of Non-suspended Benefits from SIPA, 2025

	Total	Without moratorium	With moratorium
Contributory benefits	7,258,163	2,850,565	4,407,597
Retirement pensions	5,660,776	1,777,453	3,883,323
Other pensions	1,597,387	1,073,112	524,274
Non-contributory benefits	1,458,077		

Source: Author's elaboration based on Ministerio de Trabajo, Empleo y Seguridad Social (2025).

Figure 13: Total SIPA Benefits Paid by Type of Benefit, 2001–2025



Note: Numbers correspond to monthly averages of benefits.
Source: Author's elaboration based on Ministerio de Trabajo, Empleo y Seguridad Social (2025).

Table 8: SIPA Retirement Pensions by Regime, 2025

	Number	Share of total (%)	Average benefit (pesos)
Laws prior to Law 24,241	102,298	1.80	795,092
General regime (Law 24,241)			
Without moratorium			
Standard benefits	1,410,886	24.84	880,059
Early retirement benefit (Decree 674/21)	13,950	0.25	885,047
With moratorium	3,884,970	68.40	416,436
Special regimes			
Non-university teachers	182,456	3.21	2,139,169
Energy sector workers (*Luz y Fuerza*)	17,311	0.30	2,181,064
Scientific and technological researchers	9,501	0.17	2,950,684
Judiciary	6,140	0.11	7,315,672
National university teachers	11,596	0.20	2,424,159
Foreign service	467	0.01	12,484,877
YCF Río Turbio	693	0.01	2,008,877
Provincial police and prison serv.	39,515	0.70	1,617,082
Total	5,679,693	100.00	625,510

Source: Author's elaboration based on Ministerio de Trabajo, Empleo y Seguridad Social (2025).

Law 24,241 enacted in 1993 and its subsequent amendments establish the benefits provided by the Integrated Retirement and Pension System. In the general regime, the initial benefit consists of the Universal Basic Pension (PBU), plus the Additional Benefit for Permanence (PAP). The PBU is a fixed transfer common to all beneficiaries, while the PAP depends on past contributions and is calculated as 1.5% of the average wage over the last ten years, updated based on the Pension Mobility Index, times the number of years of contributions.

When retirement is accessed through a moratorium, the benefit formula does not change, but eligibility conditions do. However, since moratoria are granted when contribution requirements are not met, the PAP is generally below the minimum pension. As a result, these benefits typically coincide with the minimum pension and are therefore independent of wages.

From 2021 to 2024, benefit adjustments were determined by the Pension Mobility Index (IMP) under Law 27,609, applied quarterly based on the growth of ANSES tax resources and wages. Let m_t denote the index applied to benefits accrued in month $t \in \{m, j, s, d\}$.[29] The IMP is

[29] $t = m$ stands for March, $t = j$ for June, $t = s$ for September, and $t = d$ for December.

calculated as follows:

$$m_t = \begin{cases} a \equiv 0.50 \times RT_t + 0.50 \times W_t & \text{if } \{t \in \{m, j, s\}\} \text{ or } \{t = d \wedge a \leq b\}, \\ b \equiv \frac{1+1.03\times\rho}{(1+m_m)(1+m_j)(1+m_s)} - 1 & \text{if } \{t = d \wedge a > b\}, \end{cases}$$

where RT_t is the variation between consecutive quarters of ANSES tax revenues in month t; W_t is the maximum between the variation of the General Wage Index (IS) published by INDEC and the variation of the Average Taxable Remuneration of Stable Workers (RIPTE) published by the Secretaría de Seguridad Social; and ρ is the variation over twelve consecutive months of total ANSES resources per benefit.

This index presents serious issues, such as not accounting for inflation risk faced by individuals and introducing lags between underlying variables and benefit adjustments. Subsequently, Decree 274/2024 modified the mobility index so that it would be based on changes in the Consumer Price Index (CPI) published by INDEC. In particular, adjustments are now monthly according to:

$$m_t = \Delta CPI_{t-2},$$

where ΔCPI_{t-2} is the monthly CPI variation corresponding to two months prior to the payment.

According to ASAP (2025), in 2024 public expenditure on social security represented around 8% of GDP, with the SIPA corresponding to over 67% of that expenditure. On the other hand, social security contributions (workers' and employers' payroll taxes) amounted to 4.4% of GDP, which corresponds to around 58.8% of total resources. The remaining resources corresponded to tax revenues channeled to ANSES, bringing total resources to approximately 7.5% of GDP. This left the social security system with an overall deficit of around 0.5% of GDP. Notably, if only payroll taxes were taken into account, the SIPA alone would have run a much bigger deficit. These ASAP figures cover the broader social-security perimeter—all ANSES revenues, including tax-earmarked transfers, against total social-security expenditure; Table 9 below instead isolates SIPA's own contributory balance, payroll contributions against contributory pension outlays alone, which is why the two deficits differ.

A.1 Expenditure as a Function of Demographics and Wages

Let a denote the age of an agent, who dies at a stochastic age, and let R denote the retirement age. Agents aged $a \leq R$ are called workers, while those aged $a > R$ are retirees. The population of age a at time t is denoted by $P_{a,t}$.[30]

[30]Notation in this appendix is local to Appendix A: $P_{a,t}$ denotes population (not the replacement ratio of Section 2), and $b^{con}_{a,t}$, $b^{mor}_{a,t}$ below denote benefit levels under the contributory and moratorium regimes (not bond holdings).

Table 9: Social Security System in Argentina (as percent of GDP), 2023-2025

	2023	2024	2025
Payroll contributions	4.3	4.4	4.8
Contributory pension outlays	5.6	5.1	5.7
Balance	–1.3	–0.7	–0.9

Source: Author's elaboration based on Subsecretaría de Seguridad Social (2024), Subsecretaría de Seguridad Social (2025) and Subsecretaría de Seguridad Social (2026).

Let $b_{a,t}^{con}$ denote the benefit received by a retiree of age a at time t without moratorium. At retirement, the benefit received is given by

$$b_{R,t}^{con} = PBU_t + \min\left\{\max\left\{\underline{b_t},\ 0.015 \cdot \overline{w}_{R,t} \cdot A_{R,t}^{con}\right\},\ \overline{b_t}\right\},$$

where $\underline{b_t}$ and $\overline{b_t}$ are the minimum and maximum benefits respectively, PBU_t is the Universal Basic Pension at time t, $\overline{w}_{R,t}$ is the average wage over the last 10 years, and $A_{R,t}^{con}$ is the number of years the agent contributed (note that $A_{R,t}^{con}$ must be 30 or greater). For simplicity, we may alternatively write $b_{R,t}^{con} = \rho(w) \cdot \overline{w}_{R,t}$, where

$$\rho(w) \equiv \frac{PBU_t + \min\left\{\max\left\{\underline{b_t},\ 0.015 \cdot \overline{w}_{R,t} \cdot A_{R,t}^{con}\right\},\ \overline{b_t}\right\}}{\overline{w}_{R,t}}.$$

Similarly, $b_{a,t}^{mor}$ denotes the benefit received by a retiree with moratorium. In order to simplify the analysis, we assume that this benefit equals the minimum pension, since this is generally the case in the data:

$$b_{R,t}^{mor} = PBU_t + \underline{b_t}.$$

Benefits are updated using the mobility index m_t, defined based on CPI variations as described above. Thus, for individuals who retire under regime $j \in \{con, mor\}$, benefits evolve as

$$b_{a,t}^{j} = (1 + m_t) \cdot b_{a-1,t-1}^{j} \quad \text{for } a > R.$$

Suppose that, in each period t, a fraction μ_t of workers aged $a = R$ retire through moratoria, while the remaining fraction $1 - \mu_t$ retire through the contributory system. Then, total pension

expenditure at time t is given by

$$G_t = \sum_{a \geq R} P_{a,t} \cdot \left[\left(1 - \mu_{t-(a-R)}\right) \cdot b_{a,t}^{con} + \mu_{t-(a-R)} \cdot b_{a,t}^{mor} \right].$$

B. The size and age distribution of the informal sector

This appendix documents the two facts about informality used in the main text: that roughly two-fifths of Argentine workers do not contribute to SIPA, and that these workers are disproportionately young. Both are computed from the microdata of the *Encuesta Permanente de Hogares* (EPH) of INDEC, pooling the four quarters of 2025 (*bases usuario individual*, T1–T4 2025), and using the survey's expansion factors (`PONDERA`). The EPH covers the 31 largest urban agglomerations, roughly 63% of the national population. All statistics refer to employed persons (`ESTADO` $= 1$), and the age distributions are restricted to ages 14–99.

Definition of an informal worker. We follow the definition of informal employment in INDEC's EPH, which classifies a worker as informal according to their occupational category (`CAT_OCUP`):

(i) *Wage earners* (`CAT_OCUP` $= 3$) are informal if their employer does not withhold pension contributions for the job (`PP07H` $= 2$, "sin descuento jubilatorio"). This is the standard measure of unregistered salaried employment in Argentina and the component most directly relevant for SIPA.

(ii) *Self-employed workers* (`CAT_OCUP` $= 2$) are informal if they either operate a business organized as a verbal partnership (`PP06A` $= 1$ and `PP06E` $= 3$) or, having no partners (`PP06A` $= 2$), work in a non-professional occupation (last digit of the CNO-2001 occupation code `PP04D_COD` greater than 1) and earn labor income in the bottom five deciles of the main-occupation income distribution (`DECOCUR` ≤ 5).

(iii) *Employers* (`CAT_OCUP` $= 1$) are informal under the same two criteria as the self-employed.

(iv) *Unpaid family workers* (`CAT_OCUP` $= 4$) are informal by construction.

A worker is formal otherwise. Note that criteria (ii)–(iii) proxy for the absence of contributions among the self-employed by the legal form and the earnings and skill level of the business, since the EPH does not ask self-employed workers whether they pay into the *monotributo* or the autonomous-worker regime.

The size of the informal sector. Table 10 reports the results. Informal workers were 41.1% of the employed population in 2025 (between 40.6 and 41.7% in the four quarters). Unregistered

wage earners account for the bulk of informal employment: 26.5% of all employed workers, or 36.7% of wage earners, do not have pension contributions withheld. Informal self-employment adds 13.1 percentage points (54% of the self-employed are classified as informal), and informal employers and unpaid family workers add 1.5 points. Informality is higher among women (43.8%) than among men (38.9%). The approximately 40% figure used in the main text (Section 4.1 and Section 6) corresponds to the total informality rate estimated here.

Table 10: Informal Employment in Argentina, 2025

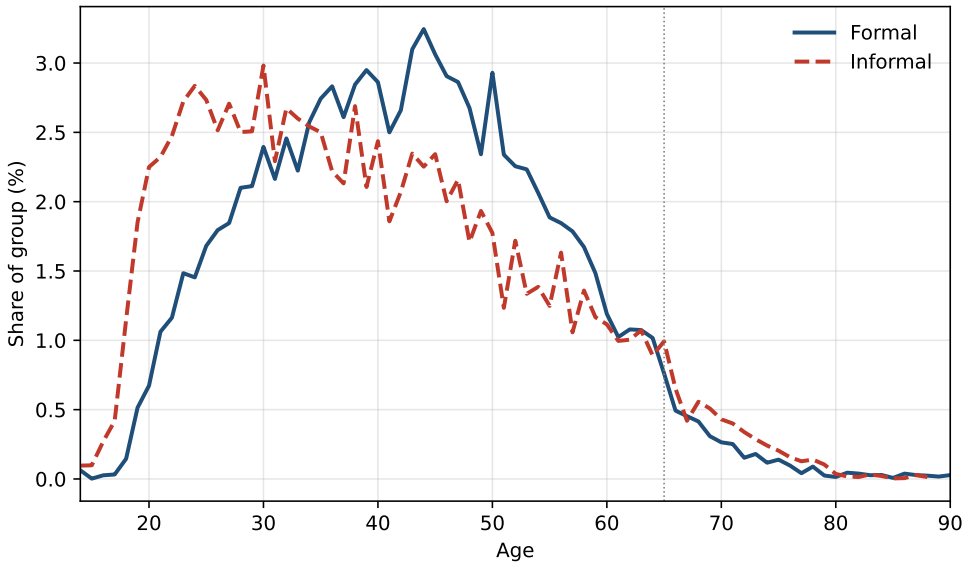
	Share of employed (%)	Share of own category (%)	Weighted average per quarter (thousands)
Wage earners without pension contributions	26.5	36.7	3,555
Informal self-employed	13.1	54.4	1,763
Informal employers	1.1	32.6	147
Unpaid family workers	0.4	100.0	53
Total informal employment	41.1		5,518
Total employment	100.0		13,418

Note: Employed persons aged 14–99 in the 31 urban agglomerations covered by the EPH, pooled over the four quarters of 2025 and weighted by `PONDERA`; counts are averages of the four quarterly weighted totals. "Share of own category" is the informal share among wage earners, the self-employed, and employers respectively. *Source*: Authors' calculations based on INDEC, Encuesta Permanente de Hogares, bases de microdatos, T1–T4 2025 (INDEC, 2025).

Age distribution. Figure 14 plots the age distribution of formal and informal workers, i.e., the share of each group at each single year of age, averaged over the four quarters of 2025. The two distributions are clearly ordered: the informal distribution peaks in the early twenties and declines from age 30 on, while the formal distribution peaks in the early-to-mid forties. The median informal worker is 38 years old; the median formal worker is 43. Table 11 shows the same fact as an informality rate by age group. Informality is U-shaped in age: 62.7% of employed 18–24 year-olds are informal, the rate falls to roughly one third between ages 40 and 54, and it rises again after 60, when the formal sector's retirement rules bind and the workers who stay employed past 65 (4.7% of the employed) are disproportionately informal. As a result, workers aged 18–29 make up 21% of employment but 29% of informal employment and only 16% of formal employment.

Implications for the model. Two consequences follow for the analysis in the main text. First, the effective contribution rate τ calibrated in Section 4.1 averages over formal and informal workers; since the informal share is highest exactly at the ages where the model's return on remaining

Figure 14: Age Distribution of Formal and Informal Workers, Argentina 2025



Note: Each line is the share of the group (formal or informal employed workers, ages 14–99) at each single year of age, computed quarter by quarter with the survey weights and averaged over the four quarters of 2025; each line integrates to 100%. The vertical dotted line marks the statutory retirement age for men (65). *Source*: Authors' calculations based on INDEC, Encuesta Permanente de Hogares, T1–T4 2025 (INDEC, 2025).

contributions $\hat{r}_{a,t}$ is lowest, the blended return at labor market entry, $\hat{r}_{0,t}$, overstates the return faced by a young worker who is formal throughout their career and pays the statutory rate. Second, the U-shape in age implies that the informal sector is not a fixed set of workers: many young workers move into formal employment in their thirties, and the pension floor delivered by the moratoria and the PUAM weakens the incentive to do so early, which is the mechanism discussed in Section 6, item (iii).

C. Population Dynamics

We derive the dynamics of population growth and the age-based demographic pyramid taking as primitives the conditional fertility and survival rates by age. Let ϕ_t and s_t be row vectors in $\mathbb{R}_+^H$ denoting fertility rates and survival probabilities by age, with $H = T + 1$ matching the main text's maximum lifespan T (Section 2). Thus, the elements of these vectors are ϕ_{ht} the number of children per woman of age h at time t, and s_{ht} the survival probability of a person of age h at time t, related to the main text's unconditional survival probabilities by $s_{h,t} = S_{h+1,t}/S_{h,t}$. The dynamics for the

Table 11: Informality and Employment by Age Group, 2025

Age group	Informality rate (%)	Share of employment (%)	Share of formal (%)	Share of informal (%)	Employment-to-population (%)
18–24	62.7	10.2	6.5	15.6	42.8
25–29	48.7	10.9	9.5	13.0	70.3
30–34	43.6	12.3	11.8	13.1	77.1
35–39	36.8	13.0	14.0	11.6	81.5
40–44	34.7	13.0	14.4	11.0	82.3
45–49	33.8	12.3	13.8	10.1	81.4
50–54	30.5	10.0	11.8	7.4	79.4
55–59	34.2	7.8	8.7	6.5	73.2
60–64	39.7	5.3	5.4	5.1	55.2
65–69	47.3	2.7	2.4	3.1	30.7
70 and over	53.1	2.0	1.6	2.6	10.4
18–64	40.5	94.9	95.9	93.4	

Note: Employed workers aged 14–99, four quarters of 2025 pooled and weighted by `PONDERA`. The informality rate is the share of employed workers in the age group who are informal under the definition above. The last column is the ratio of employed persons to the total population in the age group (all respondents in the EPH sample). Workers aged 14–17 (0.4% of employment, 90% informal) are omitted from the rows but included in the totals of Table 10. *Source*: Authors' calculations based on INDEC, Encuesta Permanente de Hogares, T1–T4 2025 (INDEC, 2025).

age distribution of the population are governed by the first-order nonlinear difference equations

$$\pi_{0,t+1} = \frac{\frac{1}{2}\phi_t \pi_t}{1 + \left(\frac{1}{2}\phi_t - (1 - s_t)\right)\pi_t} \tag{35a}$$

$$\pi_{h+1,t+1} = \frac{s_{h,t}\pi_{h,t}}{1 + \left(\frac{1}{2}\phi_t - (1 - s_t)\right)\pi_t}, \qquad \text{for } h = 0, \ldots, H-1. \tag{35b}$$

The first equation, (35a), states that the share of the population of age 0 is equal to the birth rate normalized by population growth—$\pi_{0,t+1} = \frac{\text{births}_t}{\mathscr{P}_{t+1}} = \frac{\frac{1}{2}\phi_t \pi_t \mathscr{P}_t}{\mathscr{P}_{t+1}} = \frac{\frac{1}{2}\phi_t \pi_t}{1+n_t}$. The survivors of t's generation h in the following period as a share of the $t+1$ population are given by equation (35b).

D. Certainty equivalence and the expected value of the accumulated investment

This appendix derives the accumulated value of the investment strategy used in Section 4.1, its expected value under a standard distributional assumption, and the certainty-equivalent return re-

ported there.

The accumulated investment. A household invests one unit of real cash at the start of each of T periods and reinvests dividends. Let p_t be the real price of a unit of equity and d_t the real dividend it pays. The unit invested at $t+r$ buys $1/p_{t+r}$ shares; thereafter each share pays d_{t+s}, which buys d_{t+s}/p_{t+s} additional shares, so the position grows multiplicatively and the household holds

$$n_{r,T} = \frac{1}{p_{t+r}} \prod_{s=r+1}^{T} \left(1 + \frac{d_{t+s}}{p_{t+s}}\right) \tag{36}$$

shares at $t+T$. Summing over the T contributions, wealth at the end of the accumulation phase is

$$\mathscr{A}_t = \sum_{r=0}^{T-1} \left[\prod_{s=r+1}^{T} \left(1 + \frac{d_{t+s}}{p_{t+s}}\right)\right] \frac{p_{t+T}}{p_{t+r}}. \tag{37}$$

Expression (37) simplifies considerably. Let $Q_{t+s} \equiv (p_{t+s} + d_{t+s})/p_{t+s-1}$ denote the gross real total return, and observe that

$$\left(1 + \frac{d_{t+s}}{p_{t+s}}\right) \frac{p_{t+s}}{p_{t+s-1}} = \frac{p_{t+s} + d_{t+s}}{p_{t+s-1}} = Q_{t+s}. \tag{38}$$

Writing the price ratio in (37) as $p_{t+T}/p_{t+r} = \prod_{s=r+1}^{T} p_{t+s}/p_{t+s-1}$ and pairing each factor with the corresponding dividend term using (38) gives

$$\mathscr{A}_t = \sum_{r=0}^{T-1} \prod_{s=r+1}^{T} Q_{t+s}. \tag{39}$$

Dividends and prices enter only through the total return index: the value of the strategy depends on the path of Q and not separately on the dividend yield. This is the object we compute from Shiller (2025), where $\prod_{s=r+1}^{T} Q_{t+s}$ is the ratio of the real total return index at $t+T$ to its value at $t+r$.

Expected accumulated wealth. Suppose gross real total returns are independent and lognormally distributed, $\log Q \sim \mathcal{N}(\mu, \omega^2)$. Then $\mathbb{E}\left[\prod_{s=r+1}^{T} Q_{t+s}\right] = \bar{Q}^{T-r}$, where $\bar{Q} \equiv e^{\mu+\omega^2/2} = \mathbb{E}[Q]$. Because (39) is a sum, expectations pass through term by term and

$$\mathbb{E}[\mathscr{A}_t] = \sum_{k=1}^{T} \bar{Q}^k = \bar{Q}\,\frac{\bar{Q}^T - 1}{\bar{Q} - 1}, \tag{40}$$

exactly, with no approximation. It is worth noting that the relevant object in (40) is the *arithmetic* mean gross return $\bar{Q} = e^{\mu+\omega^2/2}$ and not the geometric mean e^{μ}: expected terminal wealth exceeds

the wealth of the median path, and the gap widens with volatility and with the horizon.[31]

Certainty equivalence. Let the household consume a constant amount proportional to the wealth it has accumulated, $c = \theta \mathscr{A}_t$, throughout retirement. Under the CRRA preferences of Section 2.2, retirement utility is $\Lambda \theta^{1-\sigma} u(\mathscr{A}_t)$, where the positive constant Λ collects the discount factor and the survival probabilities. Since this is a positive multiple of $u(\mathscr{A}_t)$, the withdrawal rate θ, the discount factor and the length of retirement all cancel from the indifference condition $u(\mathscr{A}^{ce}) = \mathbb{E}\,[u(\mathscr{A}_t)]$, which yields

$$\mathscr{A}^{ce} = \left(\mathbb{E}\left[\mathscr{A}_t^{\,1-\sigma}\right]\right)^{\frac{1}{1-\sigma}} \quad \text{for } \sigma \neq 1, \qquad \mathscr{A}^{ce} = \exp(\mathbb{E}\,[\log \mathscr{A}_t]) \quad \text{for } \sigma = 1, \tag{41}$$

the geometric mean of accumulated wealth for $\sigma = 1$ and, for $\sigma = 2$,

$$\mathscr{A}^{ce} = \frac{1}{\mathbb{E}\,[1/\mathscr{A}_t]}, \tag{42}$$

the harmonic mean of accumulated wealth. The certainty-equivalent return is the constant rate that accumulates $\mathscr{A}^{ce}$ from the same contributions,

$$\sum_{k=1}^{T} (1+y^{ce})^k = (1+y^{ce})\,\frac{(1+y^{ce})^T - 1}{y^{ce}} = \mathscr{A}^{ce}, \qquad r^{ce} = (1+y^{ce})^{12} - 1. \tag{43}$$

The order of operations in (41)–(43) matters. The certainty equivalent is taken over wealth and the accumulation equation is inverted afterwards, because households consume wealth and not rates of return.

Implementation and magnitudes. We evaluate (41) under the empirical distribution of $\mathscr{A}_t$ across the overlapping 30-year windows in the sample, weighting each equally; this is the distribution whose implied returns are plotted in Figure 1b, so r^{ce} is directly comparable to the percentiles in Table 2. The resulting certainty-equivalent returns are

σ	1	2	3	5
r^{ce} (% per year)	6.8	6.3	5.9	5.1
percentile of Fig. 1b	53	43	33	20

[31] The second moment is available from the same logic. Writing $\mathscr{A}^{(n)}$ for accumulated wealth after n contributions and $m_2 \equiv \mathbb{E}[Q^2] = e^{2\mu+2\omega^2}$, iterating $\mathbb{E}\mathscr{A}^{(n)} = 1 + \bar{Q}\,\mathbb{E}\mathscr{A}^{(n-1)}$ and $\mathbb{E}(\mathscr{A}^{(n)})^2 = 1 + 2\bar{Q}\,\mathbb{E}\mathscr{A}^{(n-1)} + m_2\,\mathbb{E}(\mathscr{A}^{(n-1)})^2$ from $\mathscr{A}^{(1)} = 1$ delivers the exact mean and variance in $O(T)$ operations. Higher-order or fractional moments are not available in closed form, because $\mathscr{A}_t$ is a sum of dependent lognormal variables and is not itself lognormal.

Two remarks on robustness. First, at $\sigma = 2$ the answer does not depend on the distributional assumption: replacing the empirical distribution by the iid lognormal law of (40), calibrated to the same μ and ω, or by an iid bootstrap of historical monthly returns, changes r^{ce} by less than 0.02 percentage points. The agreement deteriorates rapidly for $\sigma \geq 5$, where the certainty equivalent is governed by the extreme left tail, on which the sample is uninformative. Second, the iid benchmark should not be read as a validation of the empirical estimate. It implies a standard deviation of $\mathscr{A}_t$ roughly three-quarters larger than the historical one, because it imposes $\sqrt{T}$ scaling on a series whose variance ratio at a 30-year horizon is about 0.36; the two calculations agree at $\sigma = 2$ because a higher mean and a fatter left tail offset one another.

E. Proofs for Section 3.3

Feasible allocations. An allocation is a non-negative array $c = \{c_{h,t}\}_{0 \leq h \leq T, t \geq 0}$ of per-capita consumption by age and date; it induces the consumption plan of every cohort. Aggregate consumption is $C_t = \sum_h \mathscr{P}_{h,t} c_{h,t}$. Households' survival-contingent claims are intermediated by competitive, actuarially fair insurers into claims on capital, on the government, and on the rest of the world, so the economy's net position vis-à-vis the rest of the world, F_t, evolves as

$$F_{t+1} = (1+r)\big(F_t + Y_t - C_t - I_t\big), \qquad I_t = K_{t+1} - (1-\delta)K_t, \tag{44}$$

in the same timing as (15a) (F_t is the amount falling due at t), together with the country's solvency condition $\lim_{t\to\infty}(1+r)^{-t}F_{t+1} = 0$, the counterpart of (15c). This is the "accumulation of net foreign assets" that aggregate consistency implies in the definition of equilibrium, written out. Iterating (44) gives $\sum_t (1+r)^{-t} C_t = F_0 + \sum_t (1+r)^{-t}(Y_t - I_t)$. With constant returns, $Y_t = W_t L_t + (r+\delta)K_t$, so $Y_t - I_t = W_t L_t + (1+r)K_t - K_{t+1}$ and $\sum_t (1+r)^{-t}(Y_t - I_t) = \sum_t (1+r)^{-t} W_t L_t + (1+r)K_0 - \lim_{T'}(1+r)^{-T'} K_{T'+1}$. By (9a), $K_t = k^*(1+\gamma)^t L_t$ and $W_t = (1+\gamma)^t\big(f(k^*) - f'(k^*)k^*\big)$, so $(1+r)^{-t}K_t$ is proportional to $(1+r)^{-t}W_t L_t$, a term of a series that converges because $L_t \leq (\max_h \theta_h)\mathscr{P}_t$ and (29) holds; the limit is therefore zero. Allowing the country to end with unused foreign wealth (free disposal), an allocation is feasible if and only if

$$\sum_{t\geq 0}(1+r)^{-t}\sum_{h=0}^{T}\mathscr{P}_{h,t}\,c_{h,t} \;\leq\; \Omega_0 \equiv \mathscr{N}_0 + \sum_{t\geq 0}(1+r)^{-t} W_t L_t, \qquad \mathscr{N}_0 \equiv F_0 + (1+r)K_0, \tag{45}$$

where $\mathscr{N}_0$, the economy's net claims on foreigners and on capital falling due at date 0, is national wealth. Since every domestic asset is a claim on capital, on the government, or on foreigners, the gross financial claims of the living, $\sum_h \mathscr{P}_{h,0}(1+r)b_{h-1,0}$, equal $\mathscr{N}_0 + \bar{B}_0$. Neither Ω_0 nor

(45) depends on the social security policy. Cohorts are the agents: with actuarially fair survival-contingent claims, consumption is certain conditional on survival, so a cohort's ex ante welfare and that of each surviving member coincide.

Proof of Lemma 3. Fix a cohort aged a at t (either $t = 0$ and $a \geq 0$, or $t = s \geq 1$ and $a = 0$). By (28) its members alive at $t+h$ number $(S_{a+h,t+h}/S_{a,t})\mathscr{P}_{a,t}$; each pays $\tau_{t+h}\theta_{a+h}W_{t+h}$ if $a+h \leq R$ and receives $p_{t+R+1-a,t+h}$ if $a+h > R$. The cohort's contribution to the primary deficit at $t+h$, discounted to date 0 at r, is therefore

$$(1+r)^{-t}\,\mathscr{P}_{a,t}\,\frac{S_{a+h,t+h}/S_{a,t}}{(1+r)^{h}}\Big[p_{t+R+1-a,t+h}\mathbf{1}_{\{a+h>R\}} - \tau_{t+h}\theta_{a+h}W_{t+h}\mathbf{1}_{\{a+h\leq R\}}\Big].$$

Summing over $h = 0,\ldots,T-a$ gives $(1+r)^{-t}\mathscr{P}_{a,t}\Psi_{a,t} = \nu_s\Psi_s$ by (26) and (27). Every unit of the deficit d_t at every date belongs to exactly one cohort, so $\sum_t (1+r)^{-t}d_t$ is the double sum over cohorts and dates of the same terms. Under (29) the double sum converges absolutely—each date's contributions are bounded by $W_tL_t \leq (\max_h \theta_h)W_t\mathscr{P}_t$ and its pensions by a multiple of $W_t\mathscr{P}_t$—so the order of summation can be exchanged, which is the identity. Iterating (15a) forward and imposing (15c) with equality gives $\sum_t (1+r)^{-t}d_t = -\bar{B}_0$, so fiscal feasibility is $\Psi(\Pi) = -\bar{B}_0$. ■

Proof of Proposition 1. *Step 1 (aggregate wealth is exhausted).* Each cohort's equilibrium plan satisfies (4) with equality, so its present value, $\sum_h \frac{S_{a+h,t+h}/S_{a,t}}{(1+r)^h}c^*_{a+h,t+h}$, equals its total wealth Ω_s. Weighting by ν_s and summing over cohorts,

$$\sum_s \nu_s\Omega_s = \underbrace{\sum_h \mathscr{P}_{h,0}(1+r)b_{h-1,0}}_{\mathscr{N}_0+\bar{B}_0} + \underbrace{\sum_s \nu_s\mathscr{W}_s}_{\sum_t(1+r)^{-t}W_tL_t} + \underbrace{\sum_s \nu_s\Psi_s}_{\Psi(\Pi)=-\bar{B}_0} = \Omega_0,$$

where the middle term allocates each date's wage bill to the cohorts that earn it (the exchange of sums as in Lemma 3) and the last uses Lemma 3 and fiscal feasibility. The equilibrium thus distributes exactly Ω_0 across cohorts, whatever Π is.

Step 2 (revealed preference). Suppose $\tilde{c}$ satisfies (45) and Pareto dominates c^*. Take any cohort s with $V_s(\tilde{c}_s) \geq V_s(c^*_s)$. If the present value of $\tilde{c}_s$ were below Ω_s, adding a small $\eta > 0$ to its consumption at its first date would be affordable and, since $u' > 0$, strictly preferred to $\tilde{c}_s$ and thus to c^*_s, contradicting the optimality of c^*_s in the cohort's budget set. So the present value of $\tilde{c}_s$ is at least Ω_s. For a cohort with $V_s(\tilde{c}_s) > V_s(c^*_s)$ the same argument gives a strict inequality: a present value at or below Ω_s would make $\tilde{c}_s$ affordable and strictly better than c^*_s.

Step 3 (summation). All terms being non-negative, sums can be exchanged, and

$$\sum_{t\geq 0}(1+r)^{-t}\sum_{h}\mathscr{P}_{h,t}\tilde{c}_{h,t}=\sum_{s}\nu_s\,\mathrm{PV}_s(\tilde{c}_s)>\sum_{s}\nu_s\Omega_s=\Omega_0,$$

where the strict inequality uses Step 2 and $\Omega_0<\infty$, so that one strict inequality added to countably many weak ones with a finite total remains strict. This contradicts (45). ■

Steps 2 and 3 are the argument of Balasko and Shell (1980, p. 295): if $\sum_t p_t\cdot x_t<\infty$ at the prices p supporting a competitive allocation x, then x is Pareto optimal, and their boundedness Property B is not needed. Their economy is a closed pure-exchange economy with date-by-date feasibility; ours is not, and the reduction that makes their argument applicable is the derivation of (45) and Step 1: production and foreign borrowing collapse into a single linear constraint at the supporting prices, and the government's lump-sum transfers, satisfying (15) at the same prices, amount to a reassignment of endowments. The commodity space is that of survival-contingent consumption, with finitely many consumers and commodities per date, the case their Section 2 covers.

Proof of Proposition 2. With the same retirement age the two policies share Ω_0 and every cohort's $b_{a-1,0}$ and $\mathscr{W}_{a,0}$. By Lemma 3 and fiscal feasibility, $\Psi(\Pi)=\Psi(\Pi')=-\bar{B}_0$; subtract. ■

Proof of Corollary 1. (a) is (30). (b) Let M be the majority, $O\neq\emptyset$ the outvoted living cohorts, and write $\nu_M\equiv\sum_{s\in M}\nu_s$, $\nu_O\equiv\sum_{s\in O}\nu_s$, and $\mathscr{B}_F\equiv\sum_{s\geq 1}\nu_s\mathscr{W}_{0,s}$, the discounted wage base of future cohorts, which is positive because $\mathscr{W}_{0,s}>0$ and finite by (29). Consider changes in social security wealth of the form

$$\Delta\Psi_s=\delta\ (s\in M),\qquad \Delta\Psi_s=-x\delta\ (s\in O),\qquad \Delta\Psi_{0,s}=-\varepsilon\mathscr{W}_{0,s}\ (s\geq 1),$$

with $(x,\varepsilon)=(\nu_M/\nu_O,0)$ for pattern (i), $(0,\delta\nu_M/\mathscr{B}_F)$ for pattern (ii), and $(2\nu_M/\nu_O,-\delta\nu_M/\mathscr{B}_F)$ for pattern (iii). In each case $\sum_s\nu_s\Delta\Psi_s=\delta\nu_M-x\delta\nu_O-\varepsilon\mathscr{B}_F=0$, and the sum converges absolutely. Implement them as follows: for a living working-age cohort $a\leq R$, keep its benefits and replace its remaining contribution rate by the flat rate $\tau'_a=\hat{\tau}_{a,0}-\Delta\Psi_a/\mathscr{W}_{a,0}$, so that $\Psi'_{a,0}=\Psi_{a,0}+\Delta\Psi_a$; for a living retiree, scale its remaining pensions by $1+\Delta\Psi_a/\mathscr{R}_{a,0}$; for the cohort entering at $s\geq 1$, replace its lifetime contribution rate by $\hat{\tau}_{0,s}+\varepsilon$, which changes its wealth by $-\varepsilon\mathscr{W}_{0,s}$. Living cohorts are finitely many and have $\mathscr{W}_{a,0}>0$ or $\mathscr{R}_{a,0}>0$, so for δ small their new rates lie in $[0,1]$ and their pensions remain non-negative; for future cohorts the new rate is $\hat{\tau}_{0,s}+\varepsilon\in[\underline{\tau}+\varepsilon,\overline{\tau}+\varepsilon]\subset[0,1]$ uniformly in s once $|\varepsilon|<\min\{\underline{\tau},1-\overline{\tau}\}$, which holds for δ small; and every cohort's total wealth stays positive (for future cohorts because $\mathscr{W}_{0,s}>0$ and the new rate is below one), so the consumption plans (7') remain well defined. The new aggregate is

$\Psi(\Pi') = \Psi(\Pi) = -\bar{B}_0$, so by Lemma 3 the new policy satisfies (15) exactly, with no residual instrument. The excluded pattern is (a). ■

F. The disutility-adjusted welfare change

This appendix derives the disutility-adjusted welfare change used in Section 4. Let

$$\widetilde{T}_{a,t} \equiv \sum_{h=0}^{T-a} \beta^h \frac{S_{a+h,t+h}}{S_{a,t}} \tag{46a}$$

$$\widetilde{T}^{w}_{a,k} \equiv \sum_{j=\max\{R+1-a,0\}}^{R+k-a} \beta^j \frac{S_{a+j,t+j}}{S_{a,t}} \tag{46b}$$

$$\omega_{a,k} \equiv \widetilde{T}^{w}_{a,k} / \widetilde{T}_{a,t} \in [0,1] \tag{46c}$$

$\widetilde{T}_{a,t}$ is the discounted, survival-weighted number of periods of life remaining at age a; $\widetilde{T}^{w}_{a,k}$ is the discounted number of *extra working* periods that raising the retirement age from R to $R+k$ imposes; and $\omega_{a,k}$ is the share of an agent's discounted remaining life that the reform converts into work. For agents already retired when the reform is announced we set $\widetilde{T}^{w}_{a,k} = 0$, so $\omega_{a,k} = 0$ and they are unaffected. Note that under retirement age R an agent works *through* age R and collects from $R+1$, so the extra periods are ages $R+1,\ldots,R+k$.

The Euler equation (5) under CRRA preferences implies $c_{a+h,t+h} = G^h c_{a,t}$ with $G \equiv (\beta(1+r))^{1/\sigma}$. Substituting into the intertemporal budget constraint (4) and solving for the initial level,

$$c_{a,t} = \frac{\mathscr{W}_{a,t}}{D_{a,t}}, \qquad D_{a,t} \equiv \sum_{h=0}^{T-a} \frac{(\beta(1+r))^{h/\sigma}}{(1+r)^h} \frac{S_{a+h,t+h}}{S_{a,t}}, \tag{47}$$

where $\mathscr{W}_{a,t} \equiv (1+r)b_{a-1,t} + W^{d}_{a,t} + R_{a,t}$ is total wealth. *At the benchmark calibration* $\beta(1+r) = 1$, so $G = 1$, consumption is constant over the life cycle at $\bar{c}_{a,t}$, and the annuity factor collapses to $D_{a,t} = \widetilde{T}_{a,t}$:

$$\bar{c}_{a,t} = \mathscr{W}_{a,t} / \widetilde{T}_{a,t}. \tag{48}$$

This is the only reason $\widetilde{T}_{a,t}$ can be read as "discounted remaining lifetime." Away from the benchmark—when r is varied while β is held fixed at its calibrated value—consumption is not flat, $D_{a,t} \neq \widetilde{T}_{a,t}$, and the expressions below must be restated in terms of $D_{a,t}$ and of $\sum_h \beta^h S_{a+h,t+h} G^{h(1-\sigma)}$ separately. We therefore report the disutility-adjusted welfare change only at the benchmark.

Discounted expected lifetime utility is simply

$$U_{a,t} \;=\; \sum_{h=0}^{T-a} \beta^h \frac{S_{a+h,t+h}}{S_{a,t}}\, u(\bar{c}_{a,t}) \;=\; u(\bar{c}_{a,t})\, \widetilde{T}_{a,t}. \tag{49}$$

Write $\widetilde{T}_a^{w,R}$ for the discounted number of working periods the agent faces under the *status quo*, i.e. up to age R. Per-period utility is $u(c) - \psi \mathbf{1}[\text{working}]$, so under the status quo

$$V_{a,t}(\Pi) \;=\; u(\bar{c}_{a,t})\, \widetilde{T}_{a,t} \;-\; \psi\, \widetilde{T}_a^{w,R} \;=\; U_{a,t} \;-\; \psi\, \widetilde{T}_a^{w,R}. \tag{50}$$

Under the reform, both consumption paths are proportional to wealth, so consumption is scaled uniformly by $\mu \equiv \mathscr{W}'_{a,t}/\mathscr{W}_{a,t} = 1 + \lambda_{a,k}$, where $\lambda_{a,k}$ is exactly the consumption-equivalent welfare change of equation (18)—an object that is *exact and independent of* σ, because it compares consumption levels rather than utilities. The agent now works through age $R+k$, so

$$V_{a,t}(\Pi') \;=\; u(\mu\, \bar{c}_{a,t})\, \widetilde{T}_{a,t} \;-\; \psi \left(\widetilde{T}_a^{w,R} + \widetilde{T}_{a,k}^{w} \right) \;=\; \mu^{1-\sigma} U_{a,t} \;-\; \psi\, \widetilde{T}_a^{w,R} \;-\; \psi\, \widetilde{T}_{a,k}^{w}. \tag{51}$$

Define $\lambda_{a,k}^{\text{net}}$ as the permanent proportional increase in status-quo consumption that leaves the agent indifferent to the reform, $V_{a,t}(c(\Pi)(1+\lambda^{\text{net}})) = V_{a,t}(\Pi')$. Scaling consumption on the left does not change how long the agent works, so by (50) and (51),

$$\left(1+\lambda_{a,k}^{\text{net}}\right)^{1-\sigma} U_{a,t} \;-\; \psi\, \widetilde{T}_a^{w,R} \;=\; \mu^{1-\sigma} U_{a,t} \;-\; \psi\, \widetilde{T}_a^{w,R} \;-\; \psi\, \widetilde{T}_{a,k}^{w}. \tag{52}$$

The baseline disutility $\psi \widetilde{T}_a^{w,R}$ is common to both sides and cancels: the reform only appends working periods at the end of the career, so the agent works the same first R periods either way. This leaves

$$\left(1+\lambda_{a,k}^{\text{net}}\right)^{1-\sigma} U_{a,t} \;=\; \left(1+\lambda_{a,k}\right)^{1-\sigma} U_{a,t} \;-\; \psi\, \widetilde{T}_{a,k}^{w}. \tag{53}$$

Two features of (53) are worth stating explicitly. First, $U_{a,t}$ appears on both sides because both sides are the *same* status-quo utility level $u(\bar{c}_{a,t})\widetilde{T}_{a,t}$, merely multiplied by different scaling factors. Second, the disutility enters additively and is *not* scaled: it is a cost in utils, not a consumption change.

Equation (32) expresses ψ as a fraction x of the flow value of certainty-equivalent consumption, $\psi = x\, u'(\bar{c}_{a,t})\, \bar{c}_{a,t}$. For CRRA preferences $u'(c)c/u(c) = 1-\sigma$, hence

$$\frac{\psi}{u(\bar{c}_{a,t})} \;=\; x(1-\sigma). \tag{54}$$

Dividing (53) through by $U_{a,t} = u(\bar{c}_{a,t})\widetilde{T}_{a,t}$ and using (54) and (46c),

$$\left(1+\lambda_{a,k}^{\text{net}}(x)\right)^{1-\sigma} \;=\; \left(1+\lambda_{a,k}\right)^{1-\sigma} \;-\; x\,(1-\sigma)\,\omega_{a,k} \tag{55}$$

For $\sigma = 1$ the limit of (55) is exact and multiplicative:

$$1+\lambda_{a,k}^{\text{net}}(x) \;=\; \left(1+\lambda_{a,k}\right)\exp\!\left(-x\,\omega_{a,k}\right). \tag{56}$$

Setting $\lambda_{a,k}^{\text{net}}(x) = 0$ in (55),

$$x^{*}(a,k) \;=\; \frac{\left(1+\lambda_{a,k}\right)^{1-\sigma}-1}{(1-\sigma)\,\omega_{a,k}}, \qquad x^{*}(a,k)\Big|_{\sigma=1} \;=\; \frac{\ln\!\left(1+\lambda_{a,k}\right)}{\omega_{a,k}}. \tag{57}$$

Agent a favours the reform if and only if $x < x^{*}(a,k)$. Setting $\sigma = 0$ in (55) and (57) recovers the linear forms $\lambda^{\text{net}} = \lambda - x\,\omega$ and $x^{*} = \lambda/\omega$; these are first-order approximations in λ and are accurate only where the consumption gain is small.